Rethinking the Arithmetic of Multi-Agent Medical Consultation

Anonymous ACL submission

Abstract

Multi-agent consultation has become the default design pattern in medical language-model systems, on the intuition that a panel of specialists diagnoses better than one physician. That intuition inherits an arithmetic premise from human medicine: pooling judgements helps because the judgements pooled are independent. We test it across 180 configurations and 63,499 episodes over three capability tiers, three benchmarks and five architectures. The premise fails twice. The specialty roster is not where the diversity comes from: of the 12.7 pp separating one doctor from the best answer anywhere in a nine-member panel, **92%** is already obtained by asking one generalist the same question nine times. And what diversity exists cannot be used. The mean pairwise error correlation is $\varphi = 0.73$, leaving a nine-member panel with 1.31 **effective independent opinions**; peer discussion raises it to 0.986, so a consultation at the moment of consensus is worth 1.01 physicians. Two specialists agree and are *both wrong* on 40.3% of items. On a scale where picking a member at random scores 0 and always picking a correct one scores 100, the five architectures score 1 to 9, because self-reported confidence separates right from wrong with Youden $J = 0.071$. Nor can independence be bought: three independently trained model families still reach only 2.6 effective opinions out of nine, and the apparent decorrelation from mixing capability is an artefact of unequal error rates. Making medical panels agree is easy. Making them *decidable* is the open problem.

1 Introduction

Ask six AI specialists a hard clinical question and let them confer. They will reach unanimity 98% of the time. They will be wrong half the times they agree. And by the moment they agree, they carry 1.01 independent opinions between them.

Now ask a different question of the same panel: was *anyone* in it right? Averaged over nine model–benchmark cells at n_a=9, some member holds the correct answer 60.7% of the time against 48.0% for one doctor working alone. The panel returns 51.2%.

A consultation is worth convening only if two things are true. Somebody in the room must be right when the others are wrong, and the room must be able to tell who. This paper measures both. Panels of language models have a little of the first and almost none of the second, and every failure reported here follows from that pair.

Multi-agent "consultation" has become the default design pattern in medical language-model systems. MedAgents (Tang et al., 2024), MDAgents (Kim et al., 2024), MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) all instantiate the same clinical intuition. A panel of specialists should diagnose better than one generalist, the reasoning goes, because a tumour board diagnoses better than one physician. The pattern is being deployed faster than it is being measured.

The intuition is not merely an analogy; it is an inheritance. Pooling diagnostic judgements improves accuracy in human medicine for a specific reason, namely that physicians trained differently make different mistakes (Kurvers et al., 2016; Kämmer et al., 2017). Majority voting converts that independence into accuracy. Strip the independence away and the arithmetic stops working, however many voters remain. Nobody has checked whether language-model panels satisfy the premise they borrow.

They do not. Measuring the pairwise error correlation between panel members across 180 multi-agent configurations, we find $\varphi = 0.73$: agents prompted into different medical specialties fail on the same items. A nine-member panel therefore carries 1.31 effective independent opinions, and peer discussion drives the correlation to 0.986, at which point a six-agent consultation is informationally equivalent to 1.01 physicians. The panel does not converge on an answer. It collapses onto one.

That is the first pillar, and it is the one prior work would predict. The second is less expected. Independence is scarce but not absent: the 12.7 pp between one doctor and the panel's oracle is real signal, held by some member on some items. On a scale where picking a panel member at random scores 0 and always picking a correct one scores 100, majority voting, an auditing orchestrator, peer debate and tiered referral score between 1 and 9. No architecture we test converts what the panel knows into what the panel says, because the quantity such a conversion needs—a signal that separates a right member from a wrong one—is missing: self-reported confidence separates them with Youden $J = 0.071$.

Three observable consequences follow, and they are the same fact seen three ways. Panel size predicts nothing, because tripling the panel adds a tenth of an opinion. Consensus carries no information, because it is agreement among copies: unanimity reaches 98% while accuracy given unanimity stays at chance, and confidence rises most on the answers that are wrong. And the benefit that survives is confined to a narrow *window* of task difficulty rather than bounded by a ceiling, roughly 25–50% single-doctor accuracy, inside which every significant improvement we measure falls and outside which every significant degradation does.

Establishing this requires a grid large enough to separate architecture from panel size from task difficulty: 180 multi-agent configurations and 63,499 episodes across three capability tiers, three medical benchmarks, five canonical architectures, panel sizes $N \in \{1, 3, 5, 7, 9\}$, and a budget-matched single-model control at every cell. Medicine makes the measurement possible in two ways that general-domain agentic benchmarks cannot. Multiple-choice clinical reasoning uses no tools, so coordination can be measured without the confound of tool-call overhead. And expert-level items drive single-agent accuracy down to 15%, far enough into the difficult regime to expose the lower edge of the window, which is invisible at benchmark difficulties where single agents already succeed. Scaling principles for general-domain agent systems (Kim et al., 2026) supply the architecture taxonomy and the controlled-comparison design our grid is built on; the error-correlation and confidence-calibration measurements that carry our argument are new here.

2 Related work

Why pooled judgements help, and what they require. Collective-intelligence results in medicine rest on error independence: pooling independent diagnostic judgements improves accuracy (Kurvers et al., 2016; Wolf et al., 2015), plateauing at roughly five raters (Kämmer et al., 2017), and groups of nine outperform individuals on the Human Dx platform (Barnett et al., 2019), the clinical instance of a general principle (Surowiecki, 2004) that multidisciplinary tumour boards institutionalise (Radcliffe et al., 2019). The framing question is not whether two heads beat one but when (Hautz and Kämmer, 2024). The same literature documents what happens when independence fails: conformity to a unanimous majority (Asch, 1956) and groupthink in deliberating groups (Janis, 1972). Correlated errors across models bound ensemble benefit in general (Kim et al., 2025), and multi-agent debate can converge on confident-but-wrong answers (Zhu et al., 2026). This paper measures that failure directly in clinical panels, where diagnostic error is already common enough in ordinary practice (Graber, 2013; Singh et al., 2014) that manufactured agreement is a safety concern rather than an efficiency one. The strongest evidence that the independence premise is what matters comes from the hybrid case: pooling 40,762 physician differentials with five frontier models over 2,133 open-ended vignettes, Zöller et al. (2025) find that human–model collectives beat physician collectives *and* model ensembles, and attribute the margin to humans and models making different kinds of errors. Their result is the consequence; the quantity that produces it is the one we measure. It also establishes the regime: independence arrives when the added judgement comes from outside the model ecosystem, which is the boundary our diversity ladder (§4.5) locates from the inside.

Medical language models and multi-agent systems. Medical question answering became a capability benchmark once models matched licensing-exam thresholds (Kung et al., 2023; Nori et al., 2023; Singhal et al., 2023, 2025), and later work moved to conducting encounters (Tu et al., 2025; McDuff et al., 2025). Prompting strategy carries much of that progress (Nori et al., 2024), which is why architectural claims need budget-matched controls. On the multi-agent side, MedAgents (Tang et al., 2024) introduced role-played specialist collaboration, MDAgents (Kim et al., 2024) added adaptive routing, and MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) formalised teamwork components, the latter sweeping team sizes 2–5. Every published medical ablation stops at $N \leq 5$, none reports a dose–response curve, and none runs a budget-matched single-model control. Skeptical evaluations find that orchestration often fails to beat well-prompted single models at substantially higher cost (Zhu et al., 2025; Liu et al., 2026; Zhao et al., 2025); our independence measurement supplies the mechanism those results lack. The pattern has since diversified along every axis except the one measured here. Chen et al. (2024b) and Zhou et al. (2025) specialise the roster—an MDT for rare disease, role-assigned radiologist and generalist agents for multimodal input—and Liu et al. (2025) adds a director to coordinate them. Ma et al. (2025) replaces majority voting with logical contradiction resolution and Li et al. (2025) lets agents accumulate diagnostic knowledge across a cohort, both of which are attempts to escape a voting rule rather than to escape the correlation that makes voting weak. Fan et al. (2024), Sanghvi et al. (2026) and Sviridov et al. (2025) move the setting itself, from single-shot selection to simulated, interactive and multimodal encounters where agents must ask before they answer, and Chen et al. (2026) and Shi et al. (2026) carry the pattern into inpatient pathways and outcome prediction, where triage, diagnosis and treatment are separate coordinated stages. These are complementary: each varies the aggregation rule, the roster or the interaction protocol, and each is a candidate for raising the capture rate we define in §4.6. None of them measures the error correlation among its own agents, so none can say whether the signal its mechanism is competing for is present.

Closest to our central question, Yuan et al. (2026) compare single-vendor and mixed-vendor panels of three frontier models on rare-disease diagnosis and report that mixed-vendor teams win, attributing the gain to complementary inductive biases surfaced by an overlap analysis. That is the same lever we isolate in Table 6, measured with a different instrument and in the regime our results predict is the favourable one—three models of comparable strength, where decorrelation carries no quality penalty. Two of our findings bear on how such comparisons should be read. Raw overlap and raw correlation are depressed mechanically when members differ in accuracy (Eq. 3), so vendor-diversity effects should be reported normalised; and within an ecosystem the lever is real but small, moving a nine-member panel from 1.31 to 2.60 effective opinions rather than towards nine.

Scaling agents, and scaling inference. Debate protocols improve factuality (Du et al., 2024; Liang et al., 2024) and frameworks such as MetaGPT (Hong et al., 2024) and AutoGen (Wu et al., 2024) made the pattern easy to deploy (Zhang et al., 2024). On how far it scales, "more agents is all you need" (Li et al., 2024) reported monotone gains, while Chen et al. (2024a) showed the relationship is non-monotone and Li et al. (2026) confirmed an inverted-U in agent count; Yang et al. (2026) attributed the shape to opinion diversity, showing two diverse agents can match sixteen homogeneous ones. Kim et al. (2026) fit a predictive scaling principle across architectures, model families and agentic benchmarks, and supply the architecture taxonomy and controlled-comparison design our grid is built on. Test-time compute offers the alternative to adding agents (Kaplan et al., 2020; Hoffmann et al., 2022; Brown et al., 2024; Snell et al., 2024), and self-consistency (Wang et al., 2023) over chain-of-thought (Wei et al., 2022) is the budget-matched control we run at every cell.

Confidence and calibration. Our safety analysis depends on whether stated confidence is informative. Models are partially aware of what they know (Kadavath et al., 2022), but elicited confidence is weakly calibrated and sensitive to how it is asked (Xiong et al., 2024; Guo et al., 2017), and clinical evaluations report fluent but unreliable responses (Goodman et al., 2023) with demographic biases that propagate (Omiye et al., 2023; Zack et al., 2024). We add a coordination-specific finding: deliberation degrades the discriminative power of confidence rather than improving it.

3 Agent systems and tasks

3.1 System definition

We describe an agent system as a triple (A, C, Ω): a set of agents A, a communication graph C, and an aggregation rule Ω. Figure 1 and Table 1 give the five configurations we evaluate, each paired with the clinical role it instantiates. The taxon-

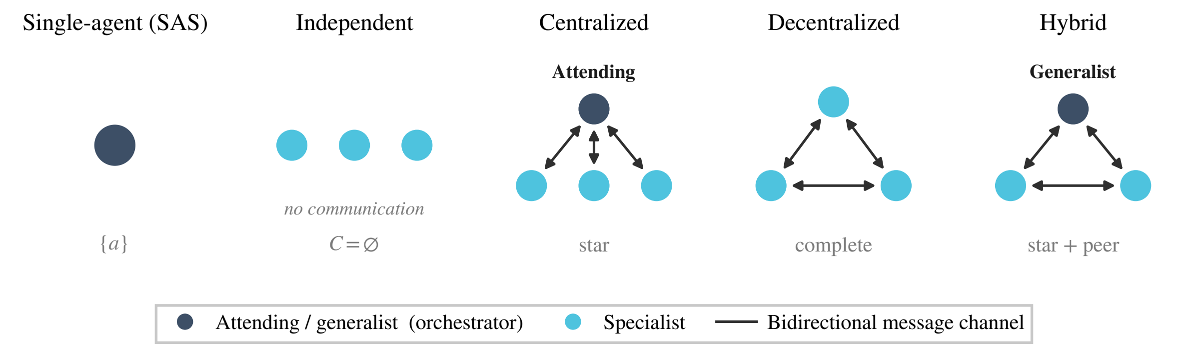

Figure 1: **The five architectures we evaluate**, with the clinical role each instantiates. A is the agent set, C the communication graph, Ω the aggregation rule.

omy is the one used for general-domain agent systems (Kim et al., 2026); the clinical instantiations are ours. Only the coordination logic differs across architectures. The item rendering, the answer schema, the role prompts and the decoding parameters are shared.

Independent. N specialists receive the item and answer without seeing one another. Aggregation is majority vote, ties broken by mean self-reported confidence and then by option order. A confidence-weighted variant is logged throughout.

Centralized (attending physician). An orchestrator receives the N specialist opinions on a star topology and is instructed to *audit before aggregating*. It must name the single opinion most likely to be wrong and justify that, then either commit to an answer or re-task one named specialty. A re-tasked specialist re-answers with the orchestrator's critique in hand. The star topology is preserved, since specialists never see each other. This design gives the panel a single point at which errors can be caught before they reach the output, and it is the architecture responsible for both the largest gain and the largest loss in the study.

Decentralized (MDT discussion). N specialists give first opinions, then exchange opinions on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent emitting the same *valid* option. A panel containing unparsed answers is not unanimous.

Hybrid (tiered referral). A generalist answers first with a stated confidence. Below a frozen threshold θ the case is referred to the N-specialist panel; if that panel has no majority the case escalates to full MDT discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set and then frozen.

Panel composition. A cheap router (`gpt-4.1-nano`, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty roster; a panel of size N is the first N of that ranking. Panels are therefore *nested*: the $N{=}9$ panel contains the $N{=}5$ panel, which contains the $N{=}3$ panel.

3.2 Nested panels as common random numbers

Every agent's first opinion is keyed on $(\text{item}, \text{specialty}_i, \text{seed}, i)$ and never on N or on the architecture. Consequently the first opinions of a panel of size N are a superset of those for any $N' < N$, and all four multi-agent architectures reuse the same first opinions. This is a deliberate common-random-numbers coupling. It reduces the variance of within-item contrasts between architectures and panel sizes, and it changes the estimand: we measure *accuracy as the same routed panel is extended*, not the expected accuracy of independently redrawn panels at each size. Paired McNemar tests on item-level outcomes remain valid under this coupling; regression models must cluster at the item level rather than treat configurations as independent.

3.3 What we measure

Accuracy alone cannot distinguish a panel that pools complementary evidence from one that repeats itself, so we instrument each episode with five coordination quantities and one derived quantity that the rest of the paper turns on.

Table 1: Architectures, their coordination structure, and the clinical role each instantiates. n is the panel size, k reasoning iterations, r orchestration rounds, d debate rounds.

Architecture	(A, C, Ω)	Cost	Clinical instantiation
Single-agent (SAS)	$\{a\}$	$O(k)$	One physician, zero-shot or CoT
Independent	$\{a_1..a_n\}$, $C = \emptyset$, synthesis	$O(nk)$	Specialists answer blind; majority vote
Centralized	$+a_{\text{orch}}$, star, hierarchical	$O(rnk)$	**Attending physician** audits, then decides
Decentralized	$\{a_1..a_n\}$, complete, consensus	$O(dnk)$	Multidisciplinary team discussion
Hybrid	$+a_{\text{orch}}$, star + peer, both	$O(rnk{+}pn)$	Tiered referral: triage $\rightarrow$ panel $\rightarrow$ MDT

We measure five coordination quantities. Their names are established in the agent-scaling literature (Kim et al., 2026), which reports headline values without always giving closed forms; the definitions below are the ones we compute.

- **Turns** T: reasoning–response exchanges; one model call is one turn.
- **Message density** c: peer-opinion exposures per turn. Independent has $C = \emptyset$ and therefore $c = 0$ by construction; Centralized exposes opinions only to the orchestrator; Decentralized and Hybrid expose every agent to every other each round.
- **Redundancy** R: mean pairwise Jaccard overlap of rationale tokens among first opinions. High R means the panel is restating one another rather than contributing independent information.
- **Overhead** $O\%$: token cost relative to the single-agent baseline on the same items, $(\text{tok}_{\text{MAS}} - \text{tok}_{\text{SAS}})/\text{tok}_{\text{SAS}}$.
- **Coordination efficiency** E_c: accuracy divided by the token-cost ratio to the single-agent baseline.
- **Error absorption**: $(E_{\text{SAS}} - E_{\text{MAS}})/E_{\text{SAS}}$, where E is the error rate.
- **Error amplification** A_e: the extent to which individual-agent errors reach the final output without inter-agent verification. We measure that failure directly, in which the panel *held* the correct option among its first opinions and still emitted a wrong one, normalised by the marginal probability that a single agent is wrong. $A_e > 1$ means the system destroys information it already possessed.

Effective independent opinions. Majority voting converts independent judgements into accuracy; correlated judgements it merely repeats. We therefore summarise a panel not by its size but by how many independent opinions it carries. Writing φ for the mean pairwise error correlation between panel positions, measured as the correlation of their per-item error indicators, the effective number of independent opinions in a panel of N agents is

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\varphi}. \tag{1}$$

At $\varphi = 0$ the N opinions are fully independent and $N_{\text{eff}} = N$; at $\varphi = 1$ they are one opinion copied N times and $N_{\text{eff}} = 1$. This is the standard design-effect correction for clustered observations, and it makes the arithmetic premise of consultation measurable rather than assumed. Reporting N_{eff} alongside N turns out to explain why panel size predicts so little (§4.5).

3.4 Clinical tasks and benchmarks

Three benchmarks, 250 items each, sampled with a fixed seed: MedXpertQA-Text (Zuo et al., 2025) (10 options; stratified by the 12 body systems $\times$ 3 task types), MedQA-USMLE (Jin et al., 2021) (4 options), and the contamination-screened hard split of MedAgentsBench (Tang et al., 2025) (894 items excluding the `medqa_5options` subset; the first 250 after stratified shuffling). The primary endpoint is exact match against the dataset's own gold option key. No LLM judge appears anywhere in the evaluation path.

Every item is additionally tagged easy / medium / hard by the pass rate of three mid-tier models at $k{=}5$ samples. On MedXpertQA the strata are heavily skewed hard (13 / 62 / 125 on the pilot sample), which is the intended regime.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form a capability ladder, measured rather than assumed (Table 15). The ladder is monotone on both benchmarks and spans single-doctor accuracy from 25% to 55% on MedXpertQA, straddling the previously reported 45% threshold from both sides. A task-grounded capability metric explains more variance than a generic intelligence index (Kim et al., 2026), so we take I to be each model's single-doctor chain-of-thought accuracy on MedQA.

Budget-matched controls. Every configuration is accompanied by three single-model controls. These are zero-shot, chain-of-thought, and **self-consistency at a matched budget**. The self-consistency pool is built from cached $n{=}8$ sampling calls (the API maximum), so any k is a prefix of the same samples. Nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what an efficient implementation would actually pay. Panels are compared against the self-consistency cell whose mean per-item *dollar* cost is closest, not the one with a matching sample count. Matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Scaling law. We fit the functional form used for general-domain agent systems (Kim et al., 2026),

$$\begin{aligned} P = \beta_0 &+ \beta_1 I + \beta_2 I^2 + \beta_3 \log(1{+}T_{\text{tools}}) \\ &+ \beta_4 \log(1{+}n_a) + \beta_5 \log(1{+}O\%) \\ &+ \beta_6 c + \beta_7 R + \beta_8 E_c + \beta_9 \log(1{+}A_e) \\ &+ \beta_{10} P_{\text{SA}} + \textstyle\sum_j \beta_j (\text{interactions}), \end{aligned} \quad (2)$$

with interactions $I{\times}E_c$, $A_e{\times}P_{\text{SA}}$, $R{\times}n_a$, $c{\times}I$ and $P_{\text{SA}}{\times}\log(1{+}n_a)$. Our tasks use no tools, so $T_{\text{tools}} = 0$ identically and every term containing it vanishes. We fit the $T{=}0$ slice, which is the point of the study. Standard errors are clustered by benchmark, and cross-validated R^2 is reported over five folds.

Statistics. Paired McNemar tests with exact binomial p-values on identical item sets, Holm-corrected within each family of comparisons. Wilson 95% confidence intervals for all accuracies. Power was calibrated on a 200-item pilot. Median discordance between configurations is 8.5%, giving a minimum detectable difference of 5.7 percentage points at $n{=}200$ and 2.6 points at $n{=}1000$ with 80% power. Configurations that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy. Scoring them as wrong answers would bias call-heavy architectures downward.

4.2 Main results

Across 180 multi-agent configurations and 63,499 episodes we find that collaboration in medicine is governed by task difficulty, not by panel size. Figure 3 gives the full dose–response surface; Table 2 the underlying numbers.

Collaboration pays only inside a window of task difficulty. Binning all 180 multi-agent configurations by the single-doctor baseline P_{SA} of their cell reveals a window rather than a ceiling (Fig. 2):

P_{SA}	n	mean gain	% positive
$<$ 25%	40	−0.88	45
25–35%	40	**+2.29**	**92**
35–50%	40	**+2.67**	**85**
50–70%	20	−0.88	50
$>$ 70%	40	+0.55	68

Configurations inside the 25–50% band gain +2.48 pp on average, while those outside it lose 0.31 pp (Welch $t = 8.42$, $p < 10^{-6}$, $n = 80$ vs 100). Splitting instead at the 45% figure of Kim et al. (2026) reproduces their direction but explains the data far less well (+1.36 pp below versus +0.07 pp above; $t = 3.46$, $p = 0.0007$). The previously reported ceiling is the window's *upper* edge. Its lower edge becomes visible only because expert-level medical items drive single-agent accuracy down to 15%. A window is not a weaker version of a ceiling. It identifies two distinct failure modes, and it describes the data better by a factor of 2.4 in t.

The window separates significant help from significant harm, with no overlap. Thirty of the 180 configurations differ significantly from the single doctor (McNemar $p < 0.05$), and they partition perfectly:

	sig. better	sig. worse
inside the window ($n = 80$)	**20**	**0**
outside the window ($n = 100$)	**0**	**10**

Every configuration that significantly beats the single doctor lies inside the window. Every configuration that significantly loses to it lies outside. The largest gain is the attending-physician (Centralized) panel at $N{=}3$: +7.60 pp at $P_{\text{SA}} = 41.6\%$ (`gpt-5-mini`, $p = 2 \times 10^{-5}$), with +6.40 pp

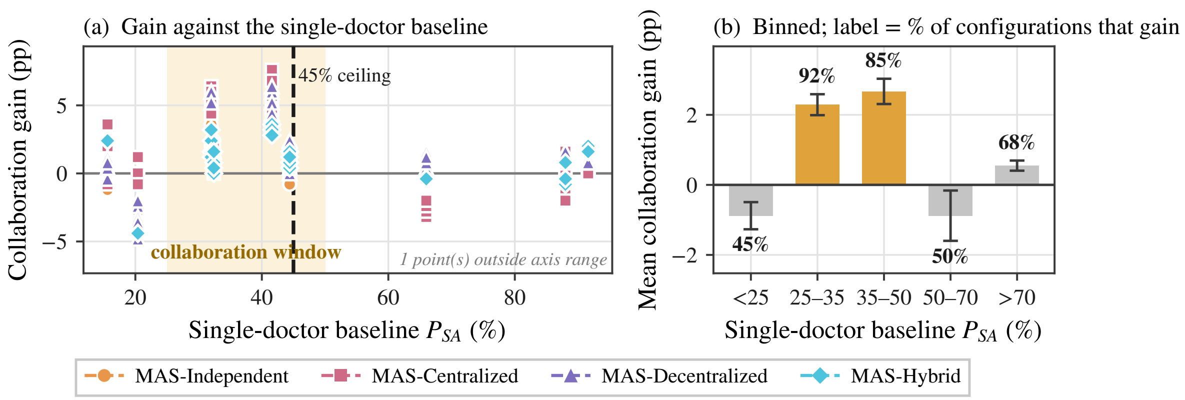

Figure 2: **Collaboration pays only inside a window of task difficulty.** (a) Collaboration gain against the single-doctor baseline P_{SA} for every multi-agent configuration; The shaded band marks the 25–50% window; the dashed line is the previously reported 45% capability ceiling. (b) The same data binned; annotations give the fraction of configurations in each bin that improve on the single doctor.

Table 2: **Main results.** For each capability tier and benchmark: the single-doctor baseline, then each architecture at its best panel size, and the gain of the best architecture over the single doctor. Bold marks the best architecture in each row. Shaded rows are the four cells whose single-doctor baseline falls inside the 25–50% collaboration window; the two largest gains in the study, +6.0 and +7.6 pp, are both shaded. The complete 180-configuration grid is Table 11.

Tier	Single doctor	Best multi-agent variant (accuracy % / n_a) Independ.	Central.	Decentr.	Hybrid	Gain best
MedXpertQA — expert-level reasoning, 10 options						
T1	15.6	16.0 / 9	**19.2** / 1	16.8 / 5	18.0 / 9	+3.6
T2	32.0	33.8 / 5	37.2 / 9	**38.0** / 7	35.2 / 5	**+6.0**
T3	41.6	47.2 / 5	**49.2** / 3	48.0 / 5	45.2 / 1	**+7.6**
MedAgentsBench-hard — items where strong models still fail						
T1	20.4	18.0 / 1	**21.6** / 7	18.4 / 5	16.0 / 9	+1.2
T2	32.4	**34.8** / 5	34.4 / 3	34.4 / 9	34.0 / 5	+2.4
T3	44.4	44.8 / 9	46.0 / 7	**46.8** / 9	46.0 / 1	+2.4
MedQA (USMLE) — licensing-exam questions, 4 options						
T1	66.0	67.2 / 9	64.0 / 9	**67.6** / 5	65.6 / 9	+1.6
T2	88.0	**89.6** / 5	**89.6** / 5	**89.6** / 9	88.8 / 5	+1.6
T3	91.6	92.4 / 1	93.2 / 1	92.8 / 5	**93.6** / 1	+2.0

■ shaded rows have a single-doctor baseline inside the 25–50% collaboration window

at $P_{\text{SA}} = 32.0\%$ (`gpt-5-nano`, $p = 0.0037$). The largest loss is also a Centralized panel, at $P_{\text{SA}} = 66.0\%$ with a weak model: -13.60 pp ($p = 10^{-5}$), where an orchestrator too weak to audit its own specialists overrides answers the panel had right. The window is not only a description of where gains concentrate. It is a two-sided boundary, and the harm on the far side is as significant as the benefit within.

The partition does not depend on where we drew the edges. Nine model–benchmark cells give nine distinct values of P_{SA} (15.6, 20.4, 32.0, 32.1, 32.4, 41.6, 44.4, 66.0, 88.0, 91.6), so the window's endpoints are identified only up to the gaps between them: any lower edge in $(20.4, 32.0]$ and any upper edge in $(44.4, 66.0]$ describes the same partition of the data. That is not a weakness to be hedged around, because the partition is *exactly invariant* over the whole identified region. Every one of $[22, 52)$, $[25, 50)$, $[25, 45)$, $[28, 48)$, $[30, 45)$, $[25, 55)$, $[25, 60)$ and $[25, 66)$ yields the identical $20/0$ inside and $0/10$ outside. The partition breaks only when the lower edge is pushed below 20.4—at $[20, 55)$ or $[15, 50)$ the count becomes $20/9$ inside—because that admits the `gpt-4.1-nano`/MedAgentsBench cell, which carries nine of the ten significant degradations. Readers asking whether 25–50% is distinguishable from 20–55% have the answer: it is, and the

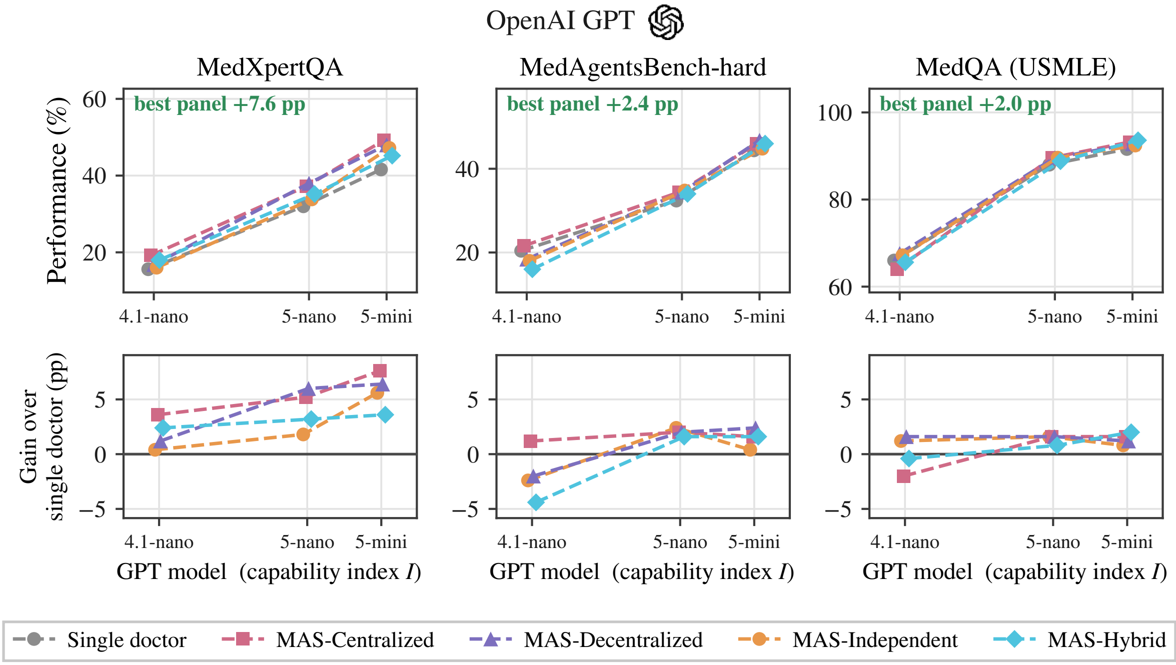

Figure 3: **Agent scaling across model capability and system architecture.** Top: performance against the task-grounded capability index I (mean single-doctor accuracy across the three benchmarks), each architecture at its best-performing panel size, grey the single doctor. Ticks mark the three GPT tiers at their measured I of 34.0, 50.8 and 59.2, and the corner figure is the best architecture's gain at the strongest model. The three benchmarks occupy different accuracy ranges, so the bottom row replots the same configurations as gain over the single doctor on one shared scale, where architecture effects are directly comparable across benchmarks. They span -4 to $+8$ pp while capability moves performance by tens of points.

reason is a single cell rather than a choice of bin width. We therefore report the window as an interval of task difficulty and not as an estimate with a spurious standard error; the boundary is where the data place it, and the data place it to within one cell.

Inside the window every architecture beats the single doctor and none beats another. Convening a panel matters; which panel does not. Taking each architecture at its best panel size in each of the four cells inside the window gives 24 pairwise architecture contrasts, and **none of them is significant** (paired McNemar, all $p \geq 0.064$, largest difference 4.0 pp). The same four cells contain 20 contrasts that significantly beat the single doctor. The decision that carries signal is whether to convene at all; the topology—star, complete, tiered, or no edges—does not separate. Panel size behaves the same way. On `gpt-5-nano` / MedXpertQA, Independent moves from 31.8% at $N{=}1$ to 32.6% at $N{=}9$, which is 0.8 pp for nine times the calls, while Decentralized reaches 38.0% at $N{=}3$ and stays there. Gains saturate at $N{=}3$ in every cell where they occur, and no architecture in any cell improves beyond $N{=}5$.

The best architecture changes with capability tier, within noise. Which architecture posts the highest point estimate does depend on the tier—at T2 it is Decentralized (38.0%, +6.0 pp) and at T3 Centralized (49.2%, +7.6 pp)—though as above the gap between them is not resolvable at this sample size. Family-dependent architectural preferences have been reported for general-domain agents (Kim et al., 2026), and the pattern here is the medical counterpart. The stronger model exploits an orchestrator's audit, whereas the weaker model benefits more from peer exchange, whose redundancy compensates for individually unreliable opinions.

4.3 Scaling principles

With accuracy as the response, the fit is degenerate. Fitting that form directly gives $R^2 = 0.997$ (5-fold cross-validated $R^2 = 0.996$), which is an artefact of our design rather than a result. The response is accuracy and one predictor is the single-doctor baseline P_{SA}; the same items and the same model underlie both terms and only the coordination structure varies, so P_{SA} alone nearly determines accuracy ($\hat{\beta}_{P_{\mathrm{SA}}} = 0.875$, $p < 10^{-4}$). We therefore refit with the collaboration *gain* as the response, which is what the scaling principle is actually about.

Coordination metrics explain the gain that accuracy alone cannot. With gain $= P - P_{\mathrm{SA}}$ as the response, $R^2 = 0.638$ and 5-fold cross-validated $R^2 = 0.415$. Coordination metrics carry that fit with no tool-count term at all. Coefficients ($n = 225$ configurations including the self-consistency control, benchmark-clustered SEs):

term	$\hat{\beta}$	p
$\log(1 + A_e)$	-0.150	< 0.001
R (redundancy)	$+0.082$	< 0.001
c (message density)	$+0.006$	< 0.001
$\log(1 + O\%)$	$+0.005$	0.024
$P_{\mathrm{SA}} \times \log(1{+}n_a)$	$+0.027$	0.048
$\log(1 + n_a)$	-0.003	0.701
P_{SA}	-0.113	0.051
P_{SA}^2	$+0.037$	0.390

Error amplification is the dominant penalty; redundancy and message density carry small positive effects; overhead is not penalised once $T{=}0$; and neither panel size nor P_{SA} has a main effect.

$T_{\mathrm{tools}} = 0$ throughout, so the $\log(1 + T)$ main effect and the $O\% \times T$ and $E_c \times T$ interactions vanish identically. Coordination metrics remain highly significant without them, which establishes that the coordination effect is not a re-description of tool overhead. The coefficient on $\log(1 + O\%)$ turns *positive* here. With no tools to coordinate over, spending more tokens on deliberation is mildly beneficial rather than costly.

The decision boundary is not identifiable in the tool-free slice. The published threshold is derived as a coefficient ratio, $P^*_{\mathrm{SA}} = \hat{\beta}_4/\hat{\beta}_{17} \approx 0.063/0.408 = 0.154$ in standardised units, or ≈ 0.45 after denormalisation (Kim et al., 2026). In our fit $\hat{\beta}_{\log(1+n_a)} = +0.0005$ ($p = 0.95$) and $\hat{\beta}_{P_{\mathrm{SA}} \times \log(1+n_a)} = +0.0074$ ($p = 0.50$). The ratio, 0.064, rests on two null coefficients and carries no information. This is a statement about functional form, not a failed replication. A linear interaction between baseline and log panel size can express a monotone ceiling but cannot express a window. The binned analysis (§4.2) shows the relationship is non-monotone in P_{SA}, and a window is what the medical data require.

Robustness of the fit. Our fit carries I and I^2 with large coefficients of opposite sign (-0.231 and $+0.228$), but bootstrap resampling ($n = 1000$) gives them standard errors of 0.94 and 0.54. With three capability tiers the quadratic is unidentifiable and we do not interpret it. Every other large coefficient is stable: $\log(1 + A_e)$ has bootstrap SE 0.024 and P_{SA} has 0.044.

Residuals pass a normality test (Shapiro–Wilk $p = 0.207$) but fail homoscedasticity decisively (Breusch–Pagan $p = 0.0001$). All standard errors in this paper are therefore heteroscedasticity-robust (HC3), clustered by benchmark where more than two clusters exist. Residual standard error is $\hat{\sigma} = 0.017$, reflecting that collaboration gains in medicine are small in absolute terms. Regularised alternatives do not improve prediction: Lasso (10-fold CV for λ) retains 9 of 11 predictors at CV $R^2 = 0.382$ and Ridge reaches 0.144, both below the unregularised 0.415, so we retain the full specification.

Nested model comparison isolates where the explanatory power lives (5-fold CV R^2): the full model reaches 0.415; capability alone 0.196; coordination metrics alone 0.093; the single-doctor baseline alone 0.043; and **panel size alone** -0.006, worse than predicting the mean. Any account of medical consultation that treats "how many agents" as the design variable is fitting noise.

The scaling principle selects an architecture, given a case mix. The fitted model converts into a design rule. For a clinical deployment characterised by its single-doctor baseline P_{SA}, expected collaboration gain is positive only for $P_{\mathrm{SA}} \in [25\%, 50\%]$, is maximised by centralized coordination at $n_a = 3$, and is negative below 25%. Three archetypes make this concrete. *Board-style examination triage* ($P_{\mathrm{SA}} \approx 0.90$, MedQA-like): use a single model; no architecture we tested returns a significant gain and every panel costs 2–5× more. *Specialist referral review* ($P_{\mathrm{SA}} \approx 0.35$, MedXpertQA-like): use an attending-physician panel at $n_a = 3$; expected gain +6 to +8 pp at a 6.7× cost premium. *Frontier diagnostic difficulty* ($P_{\mathrm{SA}} < 0.25$, MedAgentsBench-like at weak tiers): use a single model and spend the budget on a stronger one; panels lose 2.8 pp here because a panel of unreliable specialists reinforces its own errors.

Error amplification, not panel size, dominates the fit. $\log(1 + A_e)$ carries the largest coefficient of any coordination metric (-0.150, $p < 10^{-4}$). A_e measures how often a panel held the correct option among its first opinions and still emitted a wrong one. Configurations that discard information they already possess are exactly the configurations that fail to beat the single doctor. The same mechanism has been proposed from the opposite direction for general-domain agents (Kim et al., 2026); our measurement supplies the configuration-level evidence.

$R \times n_a$ is positive ($\hat{\beta} = +0.006$, $p < 0.001$), and resolving R against the ground truth says what it is measuring. Mean rationale overlap is 0.426 for pairs that are both right, 0.377 for pairs that are both wrong and 0.227 for pairs that split ($t = 95.0$ between the first two, $p < 10^{-300}$). R therefore separates *agreeing* from *disagreeing* by 0.15–0.20 and agreeing-and-right from agreeing-and-wrong by 0.049: it is a detector of whether two specialists took the same path, and only weakly of whether the path was correct. Its positive coefficient is best read as restatement tracking the items where the model is on familiar ground, not as deliberative verification.

4.4 Coordination efficiency, error dynamics, and information transfer

Coordination is cheap in turns and uninformative in messages. Fitting $T = a(n + 0.5)^b$ gives $a = 0.85$, $b = 0.770$, $R^2 = 0.299$. The exponent is sub-linear, so turns grow more slowly than agents. A clinical panel answering a closed question terminates on unanimity rather than negotiating open-ended tool use. Turn counts by architecture are Decentralized 7.5, Centralized 6.5, Independent 5.0, Hybrid 1.9, against a single-agent 1.4, a range of 1.4 to $5.3\times$. The hard resource ceiling they infer from super-linear growth does not arise in tool-free consultation.

$S = 0.462 - 0.045 \ln(c)$, $R^2 = 0.056$: message density explains almost none of the variance in accuracy. Our Centralized panels run at $c = 0.99$ and Decentralized at $c = 0.79$, well past the $c^* = 0.39$ plateau reported elsewhere, with no accuracy penalty. Once a panel is communicating at all, communicating more neither helps nor hurts.

Absorption $(E_{\mathrm{SAS}} - E_{\mathrm{MAS}})/E_{\mathrm{SAS}}$ is positive for every architecture: Decentralized $+4.2\%$, Hybrid $+2.7\%$, Independent $+1.9\%$, Centralized $+1.6\%$. Absorption is small in absolute terms and the ordering is not the expected one: peer discussion absorbs most, orchestration least.

Coordination failure is structurally impossible without a channel. We classify first-round errors into four categories, following the multi-agent failure taxonomy of Cemri et al. (2025):

	Indep.	Centr.	Decentr.	Hybrid
Logical contradiction	6.3%	6.3%	6.3%	3.4%
Numerical drift	4.5%	4.7%	4.8%	3.1%
Context omission	13.0%	13.3%	13.4%	6.8%
Coordination failure	**0.0%**	3.0%	1.2%	0.0%

Coordination failure is **0.0%** under Independent coordination, and necessarily so: with $C = \emptyset$ there is no channel on which it could occur. Hybrid is also at 0.0% because its triage gate declines to convene a panel on most items. The magnitudes do not replicate: peer verification is reported to cut logical contradiction from 16.8% to 11.5%, whereas our first-round contradiction rate is flat at 6.3% across communicating and non-communicating architectures. Numerical drift is an order of magnitude rarer than in tool-using tasks (3–5% against 19–26%), as expected when no arithmetic is cascaded.

Information gain is positive but does not buy accuracy. Information gain, the reduction in posterior variance of the success indicator between the first and final round (§D), is $+0.90$ for Decentralized, $+1.46$ for Hybrid and $+0.24$ for Centralized. Panels converge, and they converge most where the architecture communicates most. Convergence occurs whether or not the consensus is correct: $\Delta\mathcal{I}$ does not predict collaboration gain ($r = 0.055$, $p = 0.59$). Medical consultation is information-limited rather than information-transferring.

Architecture rankings do not generalise across benchmarks. Kendall's τ between the architecture rankings of any two benchmarks averages $+0.067$ (pairwise -0.200, $+0.400$, 0.000; none significant). Leave-one-benchmark-out cross-validation gives $R^2 = -0.015$: a model fitted on two medical benchmarks predicts the third no better than the mean. Coordination principles that transcend task structure in general-domain agentic work do not transcend benchmark in medicine. This is the practical counterpart of the window: architecture must be selected per difficulty regime, and difficulty regime does not coincide with domain.

The architecture that wins on accuracy is the one that costs most. Successes per 1,000 tokens: Hybrid 0.80, self-consistency 0.47, Independent

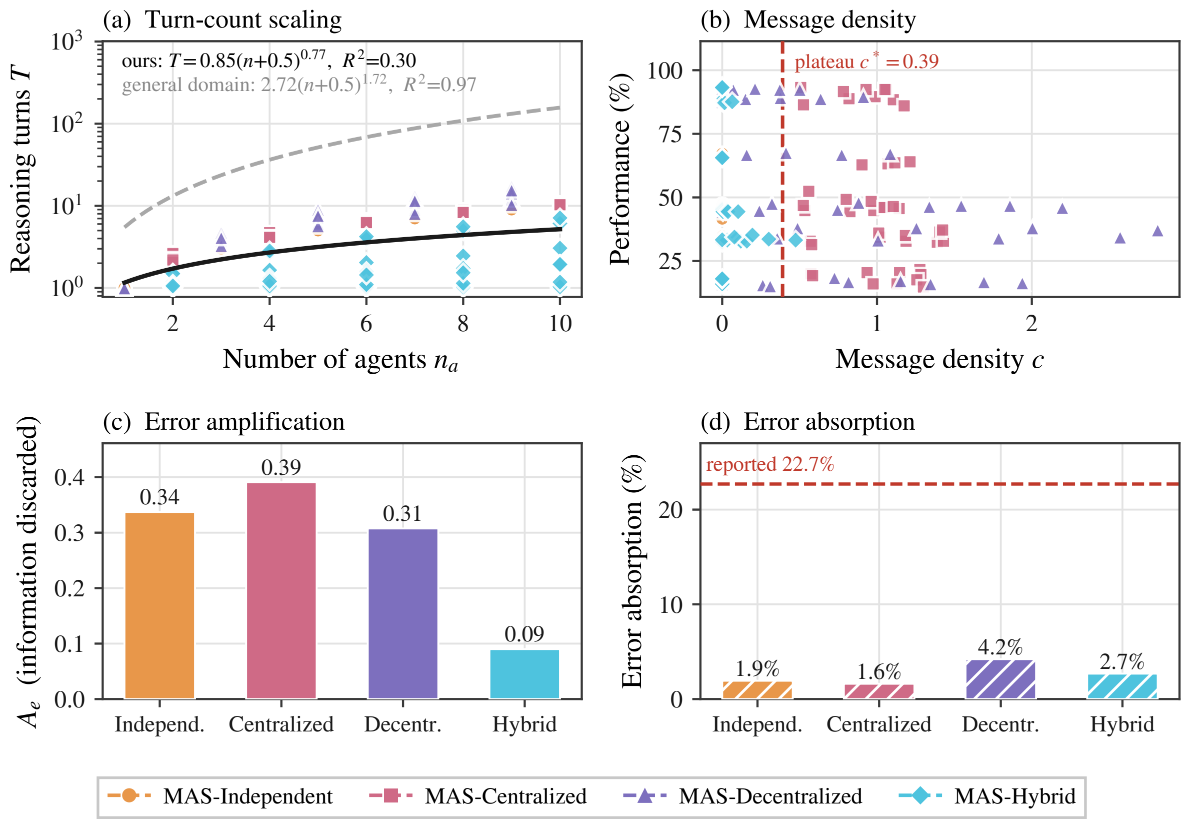

Figure 4: **Coordination dynamics.** (a) Turn count against agent count, with our fit and the published general-domain power law for comparison; medical panels converse an order of magnitude less than that law predicts. (b) Performance against message density, with a reference plateau marked at $c^* = 0.39$; performance is uncorrelated with how much the panel says. (c) Error amplification A_e, the share of information a round discards, by architecture. (d) Error absorption, the share of individually wrong opinions the panel repairs, against the published 22.7% reference.

0.39, Decentralized 0.33, Centralized 0.19. Centralized is $4.3\times$ less token-efficient than the cheapest configuration, and the architecture responsible for every significant gain is the one that costs most per correct answer. Hybrid is the most efficient, because its triage gate declines to escalate most cases and therefore rarely pays for a panel at all.

On MedXpertQA at `gpt-5-nano`, Decentralized at $N{=}3$ reaches 38.0% while self-consistency at the closest matched dollar cost reaches 36.0%. Outside the window the control wins or ties: on MedQA at the same tier, self-consistency at $k{=}15$ (88.4%) is within noise of the best panel (89.6%) at lower cost, and on MedAgentsBench at T1 self-consistency (20.0%) beats every architecture. Coordination buys something beyond repetition, and only where the window says it should.

What a point of accuracy costs. Expressing each cell's best configuration as accuracy bought per dollar of additional spend makes the deployment rule concrete (Table 3). The premium ranges from $1.9\times$ to $11.5\times$ the single doctor, and the return on it varies by two orders of magnitude: 8.5 and 4.3 percentage points per additional dollar per thousand questions on the two MedXpertQA cells inside the window, against 0.34 on MedAgentsBench at T3, where a $+2.4$ pp gain costs $11.5\times$. On MedQA at T3 the best configuration is Hybrid at $n_a{=}1$, which is *cheaper* than the single-doctor baseline because its gate almost never escalates. The window is a statement about where the money goes as much as where the accuracy does.

4.5 The arithmetic premise of consultation

Majority voting has value only when the votes are independent. Every result above is a consequence of how badly that premise fails.

A nine-member panel carries 1.3 independent opinions. Across the three architectures that convene a full panel on every item at $n_a \geq 3$ —

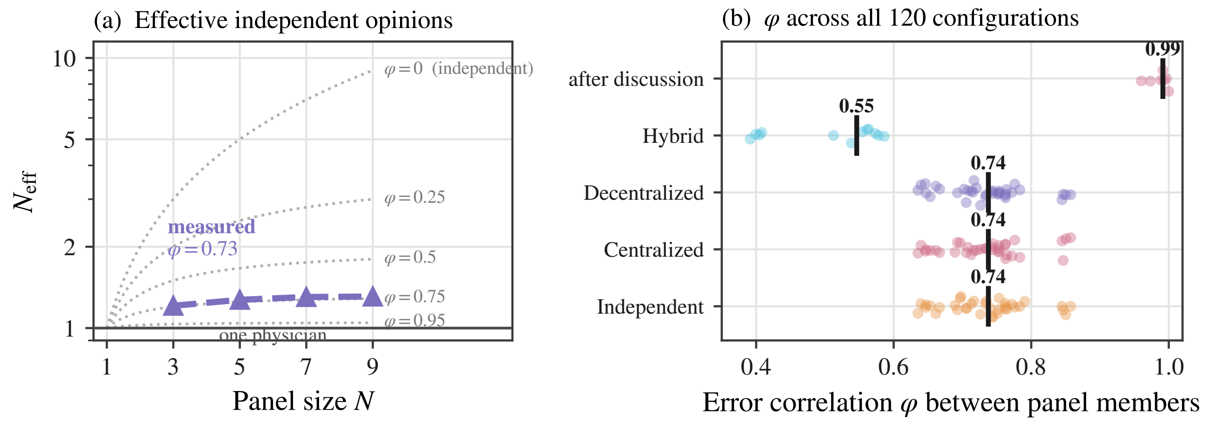

Figure 5: **Adding agents adds almost no information, and discussion removes what is left.** (a) Effective independent opinions $N_{\text{eff}} = N/(1 + (N-1)\varphi)$ against panel size, on a logarithmic scale. Dotted curves are contours of fixed error correlation φ; the solid line at $N_{\text{eff}}=1$ is a panel worth one physician. Measured panels track the $\varphi\approx0.75$ contour, so nine members carry barely more than one opinion. (b) φ for every one of the 120 grid configurations, one point per configuration, with the median marked. The value is a near-constant of the setting rather than an artefact of any one cell, Hybrid routing is the only architecture that lowers it, and peer discussion drives it to 0.99.

Table 3: **Accuracy per dollar.** For each cell, the best configuration of any architecture and panel size: its gain over the single doctor, the additional spend per 1,000 questions, the cost multiple, and the ratio. Shaded cells are inside the window.

Benchmark	Tier	gain	extra \$/1k	×cost	pp/\$
MedXpertQA	T1	+3.6	0.08	1.9	**44.5**
MedXpertQA	T2	+6.0	0.70	4.4	8.5
MedXpertQA	T3	+7.6	1.76	3.5	4.3
MedAgentsBench	T2	+2.4	0.28	2.5	8.5
MedAgentsBench	T3	+2.4	7.12	11.5	0.34
MedAgentsBench	T1	+1.2	0.39	6.2	3.1
MedQA	T1	+1.6	0.27	4.2	6.0
MedQA	T2	+1.6	0.49	4.1	3.3
MedQA	T3	+2.0	−0.14	0.8	—

Table 4: **Effective independent opinions.** φ is the mean pairwise error correlation between panel positions; $N_{\text{eff}} = N/(1 + (N-1)\varphi)$. Neither growing the panel nor changing how it is wired moves φ. The shaded row is the single exception: Hybrid's triage gate declines to convene a panel on most items, so its measured pairs come from the harder subset it does escalate.

Architecture	N	φ	N_{eff}	info. used
Growing one panel: tripling N buys 0.10 of an opinion				
Independent	3	0.743	1.21	40%
Independent	5	0.731	1.27	25%
Independent	9	0.734	1.31	15%
Changing the architecture at $N=9$: no effect either				
Centralized	9	0.735	1.31	15%
Decentralized	9	0.735	1.31	15%
Hybrid	9	**0.493**	**1.82**	20%

Independent, Centralized and Decentralized—the mean pairwise error correlation is $\varphi = 0.734$, and it is the same to three decimals in each of them separately. (Hybrid is excluded here and reported separately: its triage gate convenes a panel on only 12.4% of items, so its φ is measured on a harder subset.) By Eq. 1 this leaves $N_{\text{eff}} = 1.21$ at $N=3$, 1.27 at $N=5$ and 1.31 at $N=9$. Tripling the panel adds 0.10 of an independent opinion. Specialty role prompts do not make one model into several physicians; they make it into one physician wearing several labels.

Discussion drives the effective panel size to one. Peer discussion raises φ from 0.751 before the exchange to **0.986** after it (Fig. 5) (paired $t = 20.9$, $p = 8 \times 10^{-7}$). At $\varphi = 0.986$ a six-agent panel has $N_{\text{eff}} = 1.01$. A multidisciplinary consultation, at the moment it reaches consensus, is informationally equivalent to one physician, and the vote taken at that moment is a vote among copies. This is the mechanism behind every safety result in §4.6: unanimity rises to 98% because the opinions have merged, not because they have converged on truth; the error rate given unanimity stays at a coin flip because merging does not add information; and confidence rises on wrong answers because N agents agreeing looks like corroboration when it is repetition.

Table 5: **Clinical safety.** Unanimity is measured on the last round with two or more agents; $P(\text{wr.}\,|\,\text{un.})$ is the chance a fully agreeing panel is wrong; the last column is the confidence gap between correct and incorrect answers, the signal a clinician would use to discount an answer. The first row is the single doctor, for reference. Decentralized coordination buys the strongest appearance of agreement and the weakest usable signal at the same time.

Architecture	unan.	$P(\text{wr.}\,\vert\,\text{un.})$	conf. gap
Single doctor	—	—	**3.3**
Independent	66.3%	46.5%	2.8
Centralized	68.8%	44.9%	2.7
Decentralized	**98.3%**	49.9%	**1.8**
Hybrid†	44.4%	59.1%	1.7

the shaded row agrees almost always and is wrong half the time it agrees. †Hybrid is measured on the 12.4% of items its triage gate escalates (n = 1,392).

Table 6: **Three sources of panel diversity.** Normalised correlation $\varphi/\varphi_{\max}$ is the quantity that is not confounded by unequal error rates. Cross-family pairs span `gpt-4o`, `gpt-4.1` and `gpt-5`, three separately trained OpenAI families. Even the most diverse construction we can assemble carries 2.6 independent opinions out of nine.

Source of diversity	φ	$\varphi_{\max}$	$\varphi/\varphi_{\max}$	N_{eff}^{9}
One model, specialty prompts	0.734	0.958	0.766	1.31
Distinct models, one family	0.437	0.815	0.536	2.00
Distinct models, three families	0.308	0.686	**0.450**	**2.60**
fully independent	0	1	0	9.00

What discussion actually moves. The collapse is not agents falling silent. Across 18,196 (item, panel position) pairs, 38.6% of opinions change between the first round and the last, so deliberation is doing something. Where it takes them is the point. Of the changes, 34.6% go from wrong to right, 22.2% from right to wrong, and **43.2**% from one wrong answer to a different wrong one. Two-thirds of all movement ends somewhere other than the correct option. The net effect on individual opinions is mildly positive—872 more corrections than corruptions, or +4.8 pp of positions—and the net effect on the panel is not, because the same exchange that corrects an individual is what makes the others agree with him. Deliberation buys a small improvement in each member and pays for it with the independence that made pooling members worth anything.

The nested model comparison found panel size alone at cross-validated $R^2 = -0.006$. The independence measurement explains why that is not a null result but a necessary one: over $N \in \{3, \ldots, 9\}$, N_{eff} moves from 1.21 to 1.31. The regressor that matters barely varies, so the regressor we can control cannot predict. Any design effort spent on how many agents to deploy is spent on the wrong variable.

Correlation has a ceiling, and half of the reported decorrelation is that ceiling. Two binary error indicators with unequal marginals cannot reach $\varphi = 1$. Writing p and q for the two members' error rates, ordered $p \leq q$, the attainable maximum is

$$\varphi_{\max} = \sqrt{\frac{p\,(1-q)}{q\,(1-p)}}, \tag{3}$$

which falls as the members' accuracies diverge. Any panel assembled from models of different strength therefore posts a lower raw φ whether or not its members reason differently, and any analysis that reads raw agreement or overlap as evidence of diversity is partly reading Eq. 3. The correct quantity is the normalised correlation $\varphi/\varphi_{\max}$. Regressing the 45 model pairs in Table 6 on family identity and accuracy gap separates the two. On raw φ the accuracy gap dominates ($\hat{\beta} = -0.601$, 95% CI $[-1.034, -0.123]$, $R^2 = 0.341$). On $\varphi/\varphi_{\max}$ that coefficient vanishes and changes sign ($+0.156$, 95% CI $[-0.599, +0.994]$, $R^2 = 0.063$), while the family term is unmoved ($+0.093 \to +0.094$, both intervals excluding zero). Mixing capability does not decorrelate errors at all; it only lowers the ceiling the correlation could have reached. Mixing model families decorrelates genuinely, and by a little.

Independence is for sale, and the price is more than it is worth. The three constructions in Table 6 order cleanly, and the ordering is the practitioner's menu. Specialty prompts leave $\varphi/\varphi_{\max}$ at 0.766. Swapping in genuinely different checkpoints from one family reaches 0.536, and reaching across three independently trained families reaches 0.450—still a nine-member panel worth 2.6 opinions, not nine. The remaining lever is capability spread, and it is a trap. The least correlated pair in the study, `gpt-4o-mini` with `gpt-5-mini`, reaches $\varphi = 0.057$ by way of a 35-point accuracy gap: its members disagree because one of them is much worse. Convened as a panel, a mixed-capability roster reaches $N_{\text{eff}} =$

1.72, the highest we measure, and scores 40.8% on MedXpertQA—indistinguishable from the 41.6% of its strongest member working alone (§F.1). Independence can be bought, but only by seating someone whose errors differ because they are worse, and the panel then performs like the good doctor it diluted.

What independence does not explain. Error correlation accounts for the size and consensus results; it does not account for the window. φ correlates only weakly with the single-doctor baseline ($r = -0.215$, $p = 0.018$: harder items do carry more correlated errors) and not at all with collaboration gain ($r = -0.083$, $p = 0.37$), and a model containing φ alone reaches $R^2 = 0.007$. The window and the independence collapse are two separate facts about medical consultation: the first says when a panel can help, the second says why it cannot help much, and why its agreement should not be read as evidence.

4.6 Why no architecture helps: the answer is in the room

Error correlation explains why averaging fails. It does not explain the window, and it does not explain why architectures that never average—an orchestrator that audits, a debate that revises—fail in the same way. This section closes both gaps with a decomposition that separates what a panel *contains* from what aggregation *recovers*.

Panels disagree at a human rate, and the disagreement is sterile. Two specialists in the same panel give different answers on 17.3% of items before any exchange (MedXpertQA 24.1%, MedAgentsBench 19.3%, MedQA 7.4%). That is close to the 21% of second opinions that produce a distinctly different diagnosis in a large second-opinion audit (Van Such et al., 2017), so on this measure an AI panel looks about as diverse as a pair of physicians. Resolving the pair against ground truth shows the resemblance is superficial. Across all pairs, 42.4% agree and are right, **40.3**% agree and are *both wrong*, 9.6% split with exactly one right, and 7.7% split with both wrong. Of every disagreement, 44.5% is two agents reaching different wrong answers. The panel argues at roughly the human rate, but on two-fifths of items it has already agreed on the wrong diagnosis before the argument starts, and no aggregation rule and no panel size can reach those items.

Ninety-two percent of the panel's diversity is a temperature knob. Before asking what aggregation recovers, ask where the diversity comes from. Self-consistency gives the same model the same generic prompt nine times at the same temperature; a panel gives it nine different specialty prompts. Both produce nine answers, so both admit the same oracle measurement. At n_a=9 the single doctor scores 48.0%, nine samples of one generalist put a correct answer on the table 59.7% of the time, and nine specialists put one there 60.7% of the time. Of the 12.7 pp a specialty panel appears to offer over one doctor, **11.7** pp—92%—is already available from sampling one generalist repeatedly, and the specialty roster contributes the remaining 1.0 pp (paired over the nine model–benchmark cells, $t = 1.73$, $p = 0.12$; treating each of the 27 cell–architecture units separately, $+1.02$ pp, 95% CI $[+0.35, +1.69]$, $p = 0.004$). Whether or not that increment is distinguishable from zero, its size is fixed: role prompting is 8% of the diversity a panel contains. The specialty roster is the defining feature of the systems this design pattern is built on, and it is doing 8% of the work a temperature setting already does.

The right answer is in the room far more often than the panel says it. Let the *oracle ceiling* be the accuracy of a rule that always selects a correct first-round opinion whenever any panel member holds one. It upper-bounds every aggregation rule that does not see the label. At n_a=9 the single doctor scores 48.0%, the oracle 60.7%, and the best measured architecture 50.0%. The panel holds 12.7 pp of recoverable accuracy, and no mechanism we test recovers a tenth of it. Averaged over all configurations at $n_a \geq 3$, majority vote captures 0.6% of the headroom, budget-matched self-consistency 4.7%, an auditing orchestrator 6.5%, peer debate 8.6% and tiered referral 8.8%; selecting the best architecture per cell after the fact, which is not a deployable strategy, reaches 14%. The failure is not that a panel of one model has nothing to add. It is that nothing in the panel can tell which member is the one to listen to (Fig. 6).

The window is where signal and selectability coincide. Collaboration gain factors exactly as

$$\underbrace{P - P_{\text{SA}}}_{\text{gain}} = \underbrace{(P_{\text{oracle}} - P_{\text{SA}})}_{\text{headroom}} \times \underbrace{\kappa}_{\text{capture rate}}, \quad (4)$$

where κ is the fraction of the headroom an architecture actually recovers. The factorisation

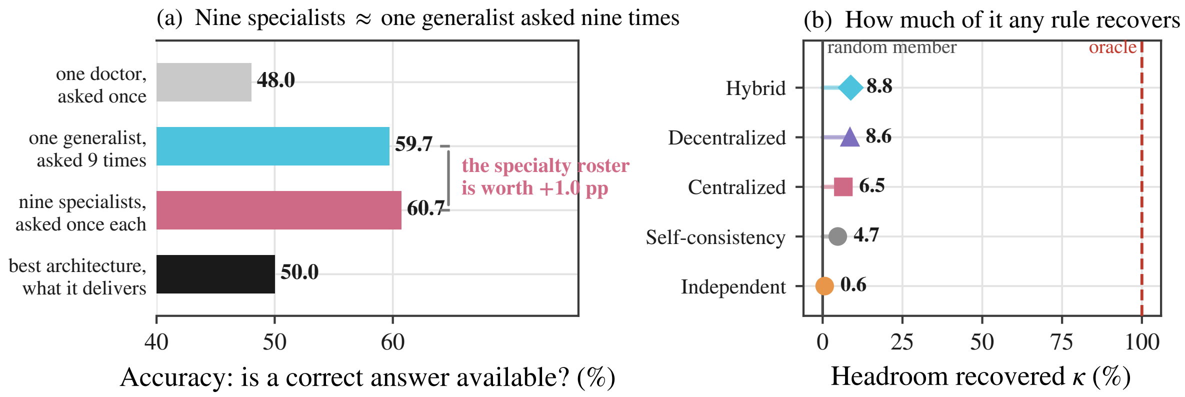

Figure 6: **A panel holds answers it cannot deliver.** (a) How often a correct answer is available at all, at n_a=9. Asking one generalist nine times makes one available 59.7% of the time; nine distinct specialists make one available 60.7%. The two middle bars are the comparison the design pattern rests on, and they are the same length: of the 12.7 pp of headroom a specialty panel appears to offer over a single attempt, 92% is sampling and 8% is the roster. The bottom bar is what the best architecture actually returns, 2.0 pp above one doctor. (b) The same gap on a selection scale, where 0 is picking one panel member at random and 100 is an oracle that always picks a correct one. Averaged over all configurations at $n_a \geq 3$, no aggregation rule reaches 9.

Table 7: **The window decomposed.** Collaboration gain is headroom times capture rate (Eq. 4). Both factors peak inside the window, and outside it at the low end aggregation actively removes accuracy the single doctor already had. n is configurations.

P_{SA} band	n	headroom	capture κ	gain
$< 25\%$	24	+6.8 pp	−21.0%	−0.96
25–50%	56	+15.5 **pp**	+17.3%	+2.55
50–70%	12	+6.7 pp	−4.4%	−0.23
$> 70\%$	27	+5.7 pp	+9.3%	+0.50

is definitional; what is not definitional is that both factors are separately measurable and that both peak inside the window (Table 7). Below $P_{\mathrm{SA}} = 25\%$ the panel holds little the single doctor misses (+6.8 pp) and aggregation *destroys* accuracy rather than recovering it ($\kappa = -21\%$). Above 70% the headroom is equally thin (+5.7 pp) and capture is weakly positive, so the net effect is nothing at all. Inside the window the headroom more than doubles (+15.5 pp) and capture turns positive (+17%) simultaneously. The window is not an artefact of where we drew the bin edges: it is the interval in which a panel both contains recoverable answers and manages not to discard them.

The headroom is item-level routing value, not a better specialty. Two oracles separate what a fixed roster could deliver from what per-item routing could. Always trusting whichever specialty slot turns out best over the whole benchmark—a post-hoc choice no deployment could make in advance—is worth 50.2% against 48.0% for the single doctor. Choosing a correct opinion item by item is worth 60.7%. Of the 12.7 pp of headroom, 2.2 is available from picking a better specialist and 10.5 requires changing which member you believe from case to case. The design question is therefore not which specialists to convene but whether anything can arbitrate between them once convened.

No signal the panel can compute tells it when it is right. Stated confidence is the obvious arbiter, and the natural objection is that a better one exists. We tested every quantity a panel can compute from its own first round: mean stated confidence, the entropy of the answer distribution, the majority share, the lowest member confidence and the spread of confidences. Against the outcome the arbiter needs to predict—whether this panel will answer correctly—all five land between AUC = 0.563 and 0.598 (Table 8), a range of 0.035. Answer entropy and majority share are marginally the best and still separate right from wrong with Youden $J = 0.185$. The problem is not the choice of signal. A panel whose members share 73% of their errors cannot distinguish the case where it agrees because the answer is easy from the case where it agrees because its members share a blind spot, and every statistic computable from its own outputs inherits that ambiguity.

The one architecture built to select cannot. Eq. 4 says the lever that matters is κ, and κ requires deciding which opinion to trust. Hybrid is

Table 8: **Every arbiter a panel could use.** Discrimination of each first-round signal against whether the panel's final answer is correct, over all 28,711 panel episodes at $n_a \geq 3$. The spread across signals is 0.035 of AUC.

First-round signal	AUC	Youden J
Answer entropy	**0.598**	**0.185**
Majority share	0.596	**0.185**
Mean stated confidence	0.593	0.127
Lowest member confidence	0.591	0.149
Confidence spread	0.563	0.118

the only architecture here that tries: a triage gate escalates to a panel when the first opinion's stated confidence falls below θ. It escalates on $\mathbf{12.4}\%$ of items overall, and the rate is set almost entirely by which model is behind the gate rather than by which items are hard—40.4% for `gpt-5-nano` on MedAgentsBench, 0.0% for `gpt-4.1-nano` on MedQA, where the gate never once fires in 1,250 items. This is not a poorly tuned threshold. It is the same measurement as §4.7 seen from the other side: a gate can only route on a signal that separates right from wrong, and stated confidence separates them with Youden $J = 0.071$. Hybrid is cheap and efficient because it is a single model most of the time, and it is a single model most of the time because there is no signal on which it could decide to be anything else.

4.7 Clinical safety: what consultation does to confidence

Accuracy is not the only thing coordination changes. In medicine the more consequential question is what it does to the *signals a clinician would act on*. Three measurements, all pointing the same way.

Self-reported confidence barely discriminates right from wrong, and discussion makes it worse. Across all configurations, mean confidence on correct answers exceeds that on wrong answers by only 3.3 points for a single doctor (86.0 vs 82.6; AUC $= 0.597$; Fig. 7). Peer discussion compresses that gap by 45%, to 1.8 points (88.4 vs 86.6; AUC $= 0.577$). The compression is not a rescaling: discussion raises confidence on *wrong* answers by 4.0 points while raising it on correct answers by 2.4. A panel that has deliberated is more sure of itself, and disproportionately so when it is mistaken. This is consistent with the threshold calibration we ran before the main experiments, where the best achievable referral gate on stated confidence reached a Youden J of only 0.071.

Discussion manufactures consensus without manufacturing correctness. Unanimity, defined as every panelist emitting the same valid option in the last round with two or more agents, rises from 66.3% under independent voting to 98.3% under peer discussion. Conditional accuracy does not follow: $P(\text{wrong} \mid \text{unanimous})$ is 46.5% for Independent and 49.9% for Decentralized. A unanimous medical panel is wrong about half the time in both cases; discussion simply produces far more unanimous panels. Centralized coordination is the only architecture that lowers the conditional error rate (44.9%), which is the error-containment role the orchestrator is supposed to play.

The most trustworthy-looking output is the least trustworthy. Taken together: consultation drives panels to near-total agreement, leaves the error rate given agreement at a coin flip, and raises confidence most on the answers that are wrong while shrinking the confidence gap that would let a reader tell the difference. The configuration a clinician would most readily act on, a unanimous and high-confidence multidisciplinary recommendation, carries essentially the same error rate as a divided one, and advertises itself more strongly. Accuracy gains of 6–8 pp inside the window are real, but they are purchased alongside a degradation in the calibration of the signal that would let a clinician catch the remaining errors.

5 Conclusion

A consultation is worth convening only if somebody in the room is right when the others are wrong, and the room can tell who. Our measurements say that panels of language models have a little of the first and almost none of the second.

The first is scarcer than the design pattern assumes, and it does not come from where the design pattern puts it. Agents prompted into different specialties make the same mistakes ($\varphi = 0.734$), leaving a nine-member panel with 1.31 effective independent opinions, and peer discussion pushes them into near-perfect agreement ($\varphi = 0.986$), leaving a six-member consultation with 1.01. Of the 12.7 pp that separates one doctor from the best answer anywhere in a nine-member panel, 92% is

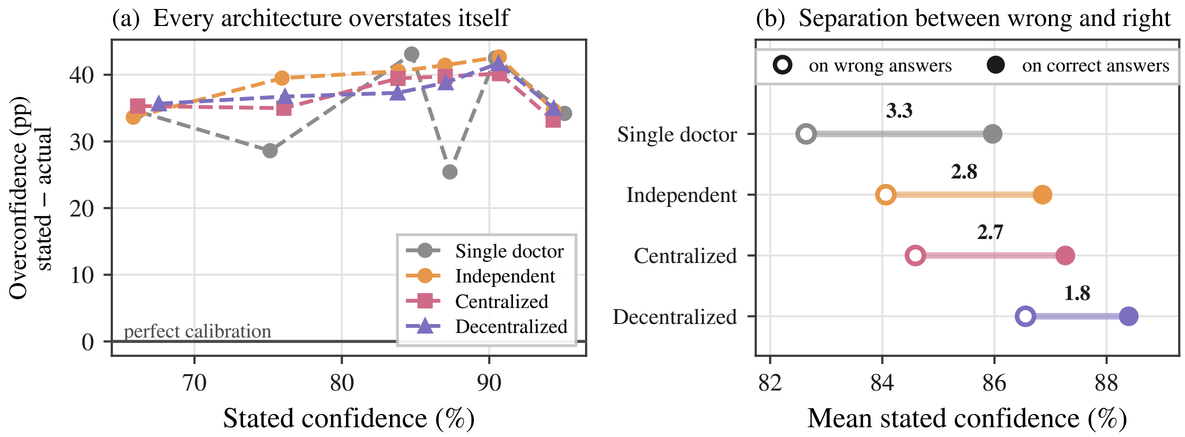

Figure 7: **Consultation degrades the signal a clinician would use to catch its errors.** (a) Overconfidence, defined as stated confidence minus the accuracy actually achieved at that confidence, against stated confidence. Perfect calibration is the solid line at zero; every architecture sits 25–43 pp above it at every confidence level, so panels are not merely uninformative about their own reliability but systematically overconfident. (b) The confidence gap between correct and incorrect answers, which is the quantity a reader would use to discount an answer. Peer discussion nearly halves it.

already available from asking one generalist the same question nine times; the specialty roster—the defining feature of the systems built on this pattern—contributes the other 8%. Independence can be bought, but not cheaply and not usefully: three independently trained model families reach 2.6 effective opinions out of nine, and the least correlated panel we can assemble reaches that state by seating a much weaker member and then performs like the stronger member working alone.

The second is the binding constraint. On a scale where picking a panel member at random scores 0 and always picking a correct one scores 100, majority voting scores 0.6, an auditing orchestrator 6.5, peer debate 8.6 and tiered referral 8.8. The panel usually holds the answer it does not give. The quantity such a system would need—a signal separating a right member from a wrong one—is missing: self-reported confidence separates them with Youden $J = 0.071$, and the one architecture here that tries to route on it convenes a panel on 12.4% of items and never once on one benchmark–model cell. Collaboration gain factors as available signal times the fraction recovered, and the window of task difficulty in which coordination pays is exactly the interval where both factors are favourable.

Three knobs the field is turning, and one it is not. Panel size: over $N \in \{3, \ldots, 9\}$ the effective number of opinions moves from 1.21 to 1.31, and panel size alone predicts gain at cross-validated $R^2 = -0.006$. The specialty roster: a temperature setting reproduces 92% of what it buys. Coordination topology: inside the window, where panels do help, all 24 pairwise architecture contrasts are non-significant, so star, complete, tiered and edgeless are indistinguishable from one another while 20 of them beat the single doctor. The knob that is not being turned is verification: every point of headroom the field is trying to reach through richer coordination protocols is already present in the panel and is being discarded for want of a way to recognise it. Where a panel is deployed anyway, the decision should be made by measuring the single-model baseline on the intended case mix first—outside the window, budget-matched self-consistency matches or beats every architecture we tested at a fraction of the cost—and capability belongs on the agents doing the reasoning rather than the one supervising them, which matters 4.1× more.

The broader caution is about what a consulting system advertises. A unanimous, high-confidence multidisciplinary recommendation is the output a clinician is most likely to act on, and in these experiments it is the output whose reliability is least legible from its own signals: unanimity reaches 98% while the error rate given unanimity stays at 50%, and the confidence gap that would let a reader tell the difference is compressed by 45%. Making medical panels agree is easy. Making them *decidable* is the open problem.

6 Limitations and future works

Our tasks are multiple-choice examination items, not clinical encounters. They carry no history taking, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. The $T = 0$ property that makes the design clean also bounds what it licenses: results here do not extrapolate to tool-using clinical agents. One finding is particularly exposed to this. Message density explains almost none of the variance in performance ($R^2 = 0.056$), which we read as coordination buying little; an equally consistent reading is that a closed-option item leaves the panel little worth transmitting. The two readings separate in open-ended settings where agents must generate a differential and request evidence, and we would expect coordination metrics to matter more there. What does not depend on the format is the selection result: whatever a panel produces, something has to decide which member to believe, and the signal available for that decision is the one we measure to be uninformative.

All models come from one vendor, so the diversity ladder in Table 6 reaches across separately trained OpenAI families rather than across vendors. We can bound what that leaves open rather than speculate about it. Within this ecosystem, moving from role prompts to distinct checkpoints to three distinct families lowers the normalised correlation from 0.766 to 0.536 to 0.450, so a nine-member panel goes from 1.31 to 2.00 to 2.60 effective opinions. A cross-vendor panel would have to decorrelate roughly twice as far again as the whole within-vendor ladder does to reach even five effective opinions out of nine, and it would have to do so at matched member capability, since the one lever that moves φ further in our data does so by seating a weaker member and costs more accuracy than the added independence returns. Both conditions are testable; neither is assumed here.

The window is a property of the model–task pairing, not of individual items. Re-deriving it within each cell, using an item difficulty defined by the leave-one-out mean accuracy of the other five models and estimating baseline and gain on disjoint halves of each stratum, does not reproduce the pattern (quadratic peak at 88%, $r = +0.13$ between stratum baseline and gain). Practitioners should read the window as a statement about which deployments to convene a panel for, not as a per-item triage rule.

Panels are nested by construction (§2.2), so the reported curves estimate accuracy as a routed panel is extended, not the expected accuracy of independently redrawn panels at each size. The coupling reduces variance on the architecture contrasts we care about and does not bias within-item paired tests, but a reader interested in the independent-panel estimand should read the curves accordingly.

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists. "More experts" and "weaker experts" are therefore entangled in the specialty-role condition. We separate them with the generic-internist control, in which every panelist is an undifferentiated internist and panel size is the only thing that varies.

Difficulty strata are defined by the pass rate of mid-tier models, some of which also appear in the evaluation ladder. This inflates the level of the hard-stratum results but not the within-item slope across panel sizes, which is computed on identical items.

Ethics Statement

This work uses only public benchmark items derived from licensing examinations and published case collections. No patient data, no human subjects and no clinical deployment are involved.

The findings carry a deployment risk that we want to state plainly rather than leave implicit. We show that multi-agent consultation systems manufacture the *appearance* of corroboration without the substance: peer discussion drives unanimity from 66% to 98% while the error rate given unanimity stays at 50%, confidence rises more on wrong answers than on right ones, and the confidence gap a reader would use to discount an answer is compressed by 45%. A unanimous, high-confidence multidisciplinary recommendation is simultaneously the output a clinician is most likely to act on and, among the outputs we measure, the one whose reliability is least legible from its own signals. Anyone deploying such a system owes its users an explicit statement that panel agreement is not evidence of correctness, and should not surface consensus or confidence as a trust cue in a clinical interface. The same property makes these systems attractive for producing medical misinformation that reads as authoritative, since apparent expert consensus is cheap to manufacture and, as we show, uninformative.

Our measurements point to three specific mitigations rather than a general caution. First, do not surface unanimity or aggregate confidence as a trust cue: in our data a unanimous panel is wrong 50% of the time, and no statistic a panel can compute about itself—stated confidence, answer entropy, majority share, minimum member confidence—separates its right answers from its wrong ones better than $\text{AUC} = 0.60$. An interface that reports "all five specialists agree" is reporting a quantity with almost no diagnostic content while implying the opposite. Second, surface the panel's *disagreements* instead, and surface them unresolved: on 9.6% of items exactly one member is right, which is where the panel's whole value lies, and a resolved consensus destroys precisely that information before the clinician sees it. Third, route verification outward. The one intervention our diversity ladder shows to be large is a judgement from outside the model ecosystem, consistent with the hybrid human–model collectives of Zöller et al. (2025); verification against retrieved evidence or a clinician is a different kind of signal from more agreement among models, and only the different kind changes the arithmetic.

We report the failure mode as a measured property of systems already being built and deployed, not as a novel capability.

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A Model capability index

Prior work scores models on a published Intelligence Index and reports that a *task-grounded* capability metric explains more variance ($R^2 = 0.413$) than the generic one ($R^2 = 0.373$). We therefore use the task-grounded form throughout: I is a model's mean single-doctor chain-of-thought accuracy across the three benchmarks. Table 9 gives the components. The three tiers span $I = 34.0$ to 59.2, and their single-benchmark baselines span 15.6% to 91.6%, crossing the 45% region from both sides. That crossing is what makes the capability-ceiling test possible at all.

B Domain complexity

B.1 Complexity metric construction

Domain complexity $D \in [0, 1]$ is the arithmetic mean of three complementary measures (Kim et al., 2026): the **performance ceiling** $1 - p_{\max}$, where $p_{\max}$ is the highest performance any evaluated system reaches; the **coefficient of variation** σ/μ of performance across all configurations, a scale-invariant measure of relative spread; and the **best-model baseline** $1 - p_{\text{best}}$, where p_{best} is the strongest single-model performance on the dataset.

B.2 Domain characterisation

Table 10 gives the components. MedQA is a low-complexity domain ($D = 0.097$); MedXpertQA and MedAgentsBench-hard are high-complexity and nearly indistinguishable ($D = 0.489$ and 0.478).

B.3 The critical threshold does not transfer

A critical complexity threshold at $D \approx 0.40$ has been reported for general-domain agents (Kim et al., 2026): below it multi-agent architectures yield net positive returns, above it coordination overhead consumes resources otherwise spent on reasoning. Our data do not support a benchmark-level threshold of any value. MedXpertQA ($D = 0.489$) and MedAgentsBench-hard ($D = 0.478$) differ by 0.011 in complexity yet by 3.5 pp in mean multi-agent gain (+3.09 pp versus −0.37 pp), giving opposite signs at effectively the same D. A benchmark-level statistic cannot separate them because the quantity that does separate them varies *within* each benchmark: on MedAgentsBench-hard the per-tier gains run −2.78, +1.02, +0.66 pp as P_{SA} moves from 20.4% through 32.4% to 44.4%, entering the window between the first tier and the second. Domain complexity is the right idea at the wrong granularity for medicine. The configuration-level baseline P_{SA} is what carries the signal.

C Datasets

MedXpertQA-Text (Zuo et al., 2025): 2,450 expert-level items, ten options each, tagged by 12 body systems and 3 task types. We draw a stratified sample interleaved across (body system × task) so that any prefix is approximately balanced, then take the first 250. **MedQA-USMLE** (Jin et al., 2021): 1,273 four-option test items; randomly shuffled with a fixed seed, first 250. **MedAgentsBench-hard** (Tang et al., 2025): the contamination-screened hard split, 894 items after excluding the `medqa_5options` subset, stratified-interleaved across the ten source subsets, first 250. All samples are nested prefixes, so a larger sample re-uses every call of a smaller one.

Item difficulty is additionally tagged easy/medium/hard by the pass rate of three mid-tier models at k=5; on the MedXpertQA sample this yields 13/62/125 items.

D Implementation details

D.1 Technical infrastructure

All models are accessed through one OpenAI-compatible client with a content-addressed disk cache, exponential backoff with rate-limit-aware retry, a per-call JSONL ledger recording tokens, latency and cost, and a cross-process spend journal guarded by an OS file lock. Every LLM call is cached on the full request (model, system, messages, temperature, seed, max tokens, effort, JSON mode, tag), so a re-run is a replay.

D.2 Agent configuration

Single agents answer in one call. Independent panels deploy N specialists with synthesis-only aggregation. Centralized systems use one orchestrator over N sub-agents with a maximum of 2 orchestration rounds and one re-task per round. Decentralized systems run N agents through up to 3 discussion rounds, stopping early only on true unanimity. Hybrid systems combine a generalist triage gate with the specialist panel and escalate to full discussion when the panel has no majority. Agent temperature is 0.7 within panels, 0.3 for single-doctor baselines and for orchestrator/moderator turns; reasoning models are called at `effort=low` with a minimum completion budget of 1,600 tokens, since reasoning tokens are drawn from the same budget and a smaller cap returns empty output.

D.3 Prompt compilation

Every agent receives the same rendered item (stem, then options as `(A) ...`) and the same answer schema: JSON with `answer`, `confidence`

Table 9: Capability index and its components (single-doctor CoT accuracy, %).

Model	MedXpertQA	MedAgentsBench	MedQA	I (mean)	Reasoning
`gpt-4.1-nano`	15.6	20.4	66.0	**34.0**	—
`gpt-5-nano`	32.0	32.4	88.0	**50.8**	effort=low
`gpt-5-mini`	41.6	44.4	91.6	**59.2**	effort=low

shaded tiers straddle the 45% ceiling reported for general-domain agents

Table 10: Domain complexity of the three medical benchmarks.

Benchmark	$1 - p_{\max}$	σ/μ	$1 - p_{\text{best}}$	D	mean MAS gain
MedXpertQA	0.508	0.376	0.584	**0.489**	+3.09 pp
MedAgentsBench-hard	0.532	0.346	0.556	**0.478**	−0.37 pp
MedQA (USMLE)	0.060	0.147	0.084	**0.097**	+0.07 pp

shaded benchmarks are the two whose difficulty places them inside the window

(0–100) and `reason` (at most 45 words). Only the system prompt differs, naming the specialty. Peer context in communicating architectures is capped at a fixed 900-character budget, balanced-truncated across peers and *independent of* N, so a panel-size effect cannot be a context-length effect. Peer ordering is deterministically rotated per (item, round) so that the first-listed speaker is decorrelated from the top-ranked specialty.

D.4 Evaluation methodology

Sample sizes. 250 items per benchmark, 180 multi-agent configurations, 63,499 episodes. **Controls.** All architectures share the item rendering, answer schema, role-prompt template and decoding parameters; only coordination logic differs. Model weights are frozen. The primary endpoint is exact match against the dataset's own gold option key. No LLM judge appears anywhere in the evaluation path. Episodes that fail for infrastructure reasons carry an error status and are excluded from accuracy rather than scored as wrong, which would bias call-heavy architectures downward. Cost accounting is nominal: cached calls are charged at list price, so reported economics describe what a configuration would cost to run.

D.5 Information gain computation

Information gain is the Bayesian posterior variance reduction (Kim et al., 2026)

$$\Delta\mathcal{I} = \tfrac{1}{2}\log\frac{\mathrm{Var}[Y \mid \mathbf{s}_{\text{pre}}]}{\mathrm{Var}[Y \mid \mathbf{s}_{\text{post}}]}, \qquad (5)$$

with $\mathrm{Var}[Y \mid \mathbf{s}] = \hat{p}(\mathbf{s})(1-\hat{p}(\mathbf{s}))$ and $Y \in \{0,1\}$ the success indicator. They estimate $\hat{p}$ from $K{=}10$ sampled reasoning traces at temperature 0.7. In multiple-choice consultation the panel already supplies such a sample: $\hat{p}$ is the fraction of the round's opinions that name the gold option, Laplace smoothed as $(k{+}1)/(n{+}2)$. Smoothing rather than clipping matters. A Centralized panel's final round contains only the orchestrator, and clipping would pin $\hat{p}$ at 0.5 and force $\Delta\mathcal{I}$ negative by construction.

E Complete results grid

F Additional results

F.1 Where capability belongs inside a panel

Specialist capability matters four times more than the orchestrator's. Replacing the whole panel with weak models costs 34.4 pp (49.2% to 14.8%). Splitting that substitution isolates where the loss comes from. Downgrading only the *orchestrator*, keeping strong specialists, costs 3.6 pp, so the panel retains 90% of the homogeneous-high span. Downgrading only the *specialists*, keeping a strong orchestrator, costs 14.8 pp, retaining 57%. A strong attending physician recovers more than half of what a weak panel loses, but a weak attending physician barely dents a strong one. The asymmetry is $4.1\times$. In a centralized clinical panel, capability should therefore be spent on the agents doing the reasoning rather than on the agent doing the auditing.

An orchestrator too weak to audit is worse than no orchestrator at all. The largest single degradation anywhere in this study is a Centralized panel with a weak model on MedQA at $P_{\text{SA}} = 66.0\%$: -13.60 pp, $p = 10^{-5}$. Inspection of those episodes shows the mechanism directly. The orchestrator overrides a correct specialist majority.

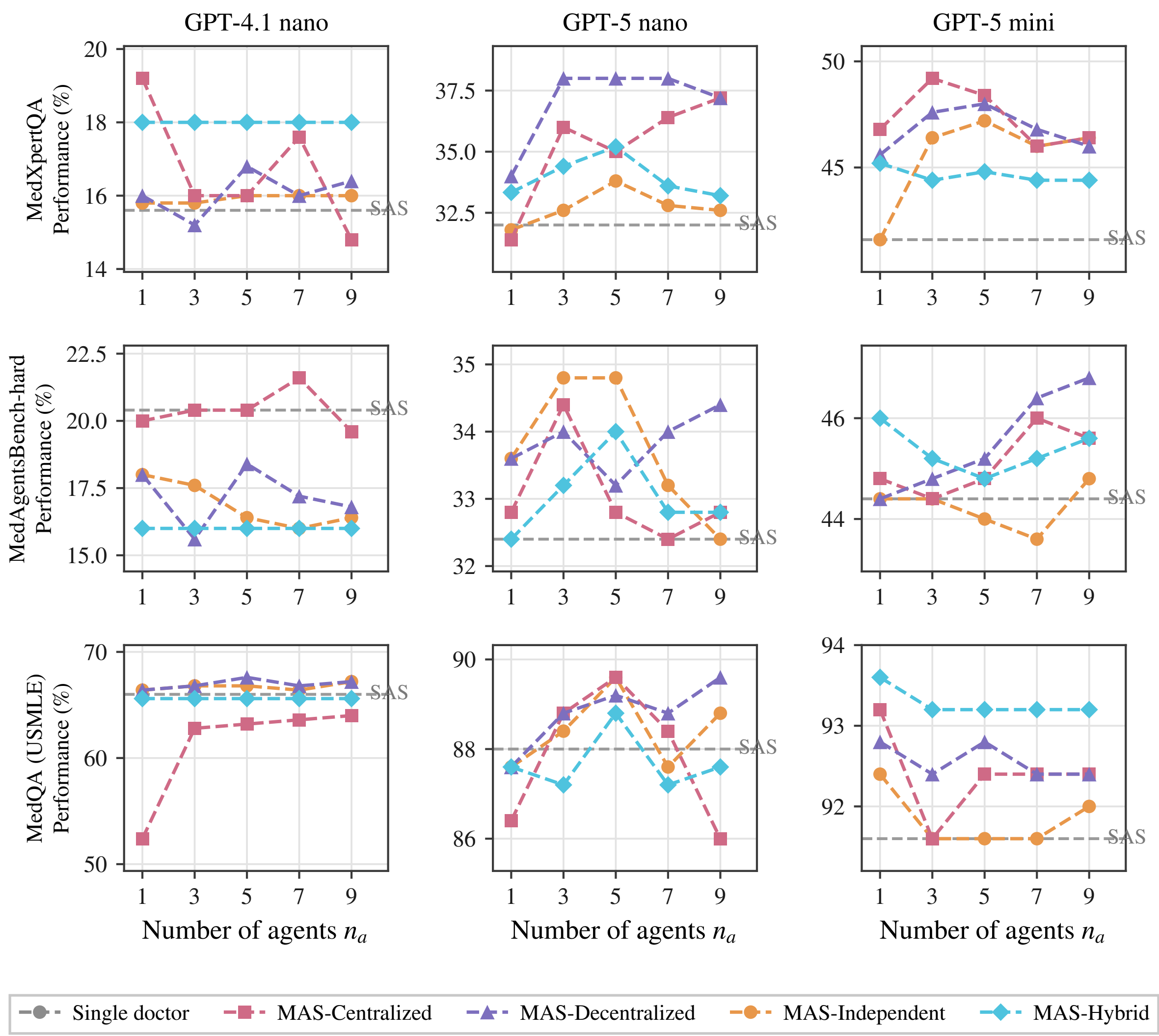

Figure 8: **Number-of-agents scaling.** Performance against panel size $n_a \in \{1, 3, 5, 7, 9\}$ for every architecture, model and benchmark. Dashed grey: single-agent baseline. Dotted: budget-matched self-consistency. Where gains occur they are captured by $n_a = 3$.

This is the error-amplification term in its most extreme form. It is also the counterpart to the previous result. The audit is what makes centralized coordination work, so an auditor that cannot audit converts the architecture's strength into its worst liability.

Mixed-capability panels land between their components, with no emergent benefit. A panel of one strong, one mid and one weak specialist reaches 40.8% under Centralized and 44.0% under Decentralized, against 49.2% and 47.6% for homogeneous-high. Both sit below the strong homogeneous panel and well above the weak one, roughly where a capability-weighted average would put them. We find no medical analogue of the heterogeneous-mixing advantage reported for one model family (Kim et al., 2026), in which a low-capability orchestrator with high-capability sub-agents outperformed a homogeneous high-capability panel by 31%. In our data that configuration is the second-best Centralized cell but still 7% below homogeneous-high, directionally matching what they observe for OpenAI models, where heterogeneous centralized configurations degrade.

F.2 Where in medicine collaboration pays

Collaboration helps diagnosis most, and it is the only task type significant at both tiers. MedXpertQA labels each item as Diagnosis, Treatment, or Basic Science. Splitting the window-resident cells by label, the best panel gains +9.5 pp on Diagnosis at `gpt-5-nano` ($p = 0.008$) and +10.7

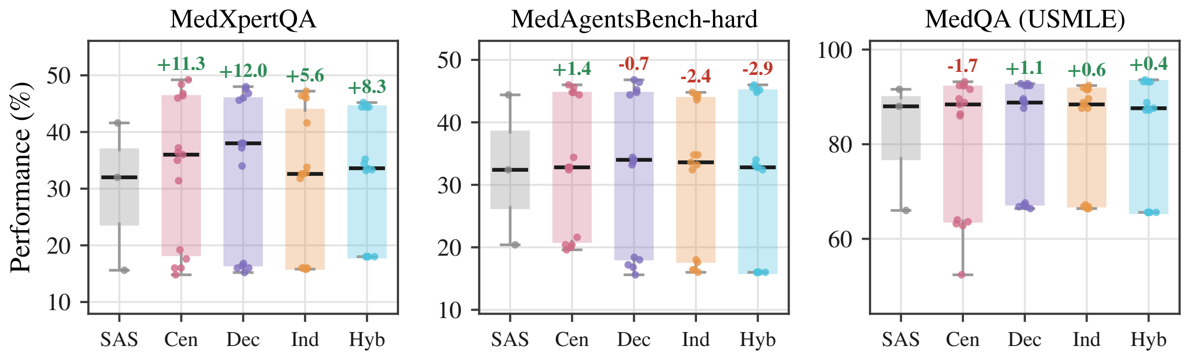

Figure 9: **Single doctor versus panel: performance distributions.** Box plots over all configurations of each architecture, with individual configurations overlaid. Bars are labelled SAS (single doctor), Cen, Dec, Ind and Hyb for Centralized, Decentralized, Independent and Hybrid. Annotations give the change in mean relative to the single doctor, as a percentage of that baseline. The spread within an architecture dwarfs the difference between architectures.

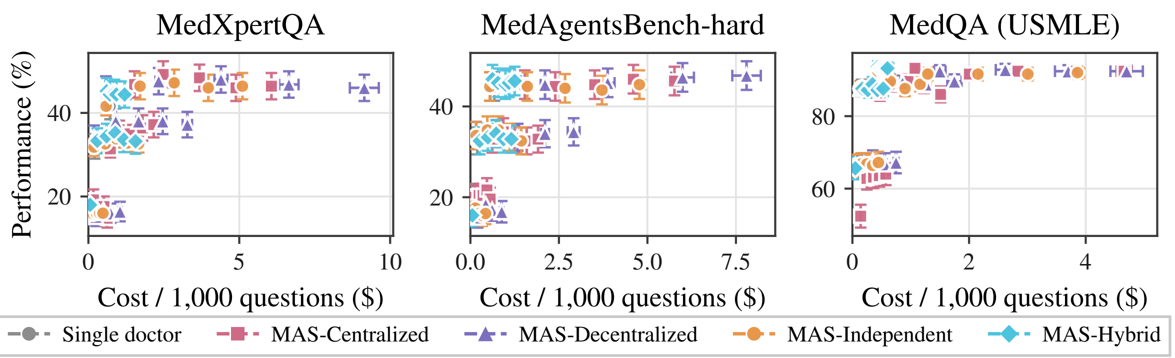

Figure 10: **Cost–performance trade-offs.** Mean performance against nominal cost per 1,000 questions, with SEM error bars on both axes. Each point is one configuration.

pp at `gpt-5-mini` ($p = 0.004$), against $+7.1$ and $+8.3$ pp on Treatment and $+7.3$ and $+8.5$ pp on Basic Science. Diagnosis is the only category significant at both tiers. This is the pattern clinical intuition predicts: differential diagnosis is the medical task that most resembles pooling independent perspectives, and it is where pooling pays.

The difficulty stratification is where the window shows up within a single benchmark. On MedXpertQA at `gpt-5-mini`, the best configuration gains $+7.7$ pp on easy items (n=13), $+6.5$ pp on medium (n=62) and $+9.6$ pp on hard (n=125). Gains are positive in every stratum for this tier, which sits inside the window; the sign flips only for tiers whose baseline falls outside it. Item difficulty and configuration baseline are two views of the same quantity. The window is defined on the view that can be measured before deployment.

G Effect sizes by configuration

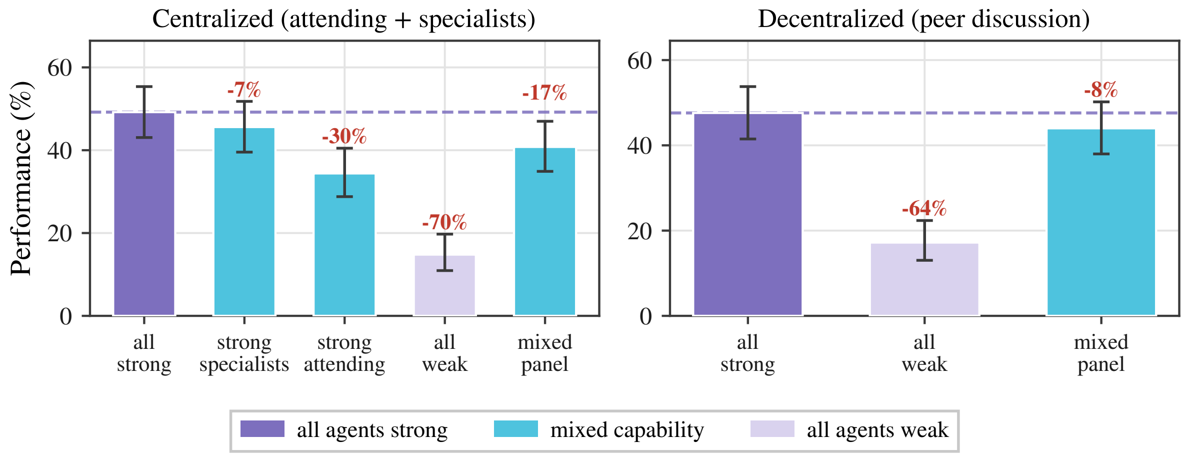

Figure 11: **Agent heterogeneity.** Performance of Centralized and Decentralized panels at n_a=3 on MedXpertQA under five capability placements. Strong is `gpt-5-mini`, weak is `gpt-4.1-nano`; *strong specialists* pairs a weak attending with strong specialists and *strong attending* reverses that pairing. Annotations give the change relative to the all-strong panel (dashed line). Decentralized coordination has no attending, so the two attending-capability conditions are structurally undefined there and are omitted.

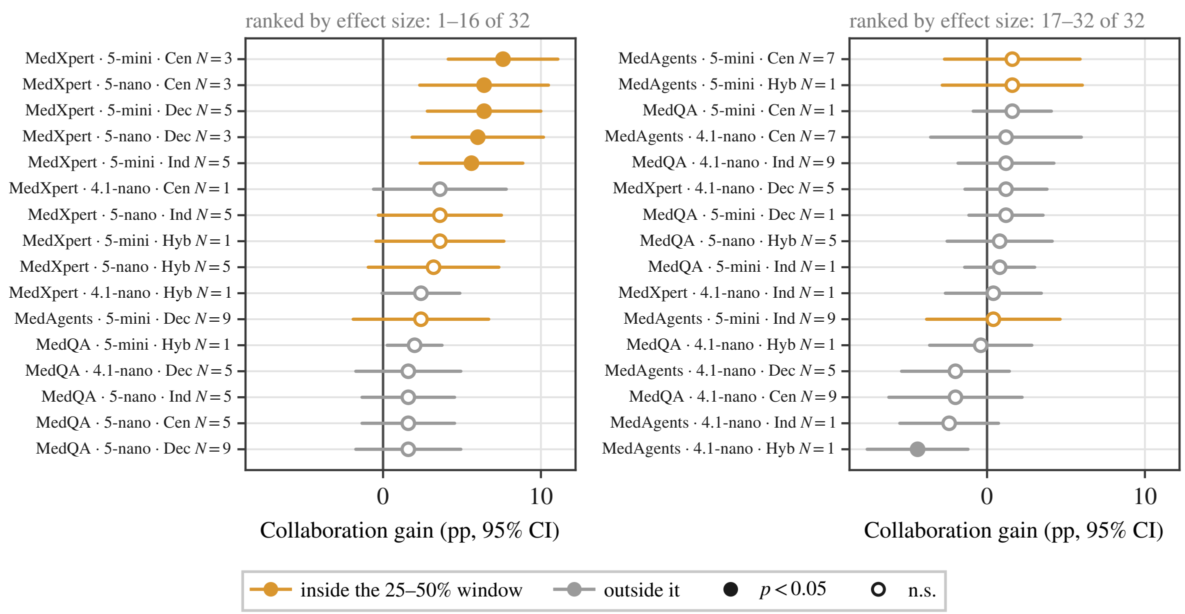

Figure 12: **Effect size for every configuration, ranked.** Collaboration gain with 95% confidence intervals; each row is one architecture at its best panel size within a difficulty stratum, and rows run left to right in descending order of effect. Amber rows have a single-doctor baseline inside the 25–50% window, grey rows outside it. Filled markers are significant at $p < 0.05$, hollow are not. The ranking is what carries the claim: amber occupies the positive tail and grey the negative one, and every significant row on the positive side is amber while the only significant row on the negative side is grey.

Table 11: Complete results grid (1 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T1	MedAgentsBench	Centralized	1	20.0	[15.5, 25.4]	0.14
T1	MedAgentsBench	Centralized	3	20.4	[15.9, 25.8]	0.25
T1	MedAgentsBench	Centralized	5	20.4	[15.9, 25.8]	0.36
T1	MedAgentsBench	Centralized	7	21.6	[16.9, 27.1]	0.47
T1	MedAgentsBench	Centralized	9	19.6	[15.2, 25.0]	0.56
T1	MedAgentsBench	SAS CoT	1	20.4	[15.9, 25.8]	0.08
T1	MedAgentsBench	Decentralized	1	18.0	[13.7, 23.2]	0.05
T1	MedAgentsBench	Decentralized	3	15.6	[11.6, 20.6]	0.21
T1	MedAgentsBench	Decentralized	5	18.4	[14.1, 23.7]	0.43
T1	MedAgentsBench	Decentralized	7	17.2	[13.0, 22.4]	0.61
T1	MedAgentsBench	Decentralized	9	16.8	[12.7, 21.9]	0.88
T1	MedAgentsBench	Independent	1	18.0	[13.7, 23.2]	0.05
T1	MedAgentsBench	Independent	3	17.6	[13.4, 22.8]	0.14
T1	MedAgentsBench	Independent	5	16.4	[12.3, 21.5]	0.24
T1	MedAgentsBench	Independent	7	16.0	[12.0, 21.1]	0.33
T1	MedAgentsBench	Independent	9	16.4	[12.3, 21.5]	0.43
T1	MedAgentsBench	Self-consist.	1	20.0	[15.5, 25.4]	0.05
T1	MedAgentsBench	Self-consist.	3	18.8	[14.4, 24.1]	0.09
T1	MedAgentsBench	Self-consist.	5	19.2	[14.8, 24.5]	0.13
T1	MedAgentsBench	Self-consist.	9	18.0	[13.7, 23.2]	0.25
T1	MedAgentsBench	Self-consist.	15	18.4	[14.1, 23.7]	0.37
T1	MedAgentsBench	Hybrid	1	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	3	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	5	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	7	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	Hybrid	9	16.0	[12.0, 21.1]	0.05
T1	MedAgentsBench	SAS zero-shot	1	16.8	[12.7, 21.9]	0.03
T1	MedQA	Centralized	1	52.4	[46.2, 58.5]	0.14
T1	MedQA	Centralized	3	62.8	[56.7, 68.6]	0.25
T1	MedQA	Centralized	5	63.2	[57.1, 68.9]	0.36
T1	MedQA	Centralized	7	63.6	[57.5, 69.3]	0.46
T1	MedQA	Centralized	9	64.0	[57.9, 69.7]	0.57
T1	MedQA	SAS CoT	1	66.0	[59.9, 71.6]	0.08
T1	MedQA	Decentralized	1	66.4	[60.3, 72.0]	0.05
T1	MedQA	Decentralized	3	66.8	[60.7, 72.3]	0.20
T1	MedQA	Decentralized	5	67.6	[61.6, 73.1]	0.35
T1	MedQA	Decentralized	7	66.8	[60.7, 72.3]	0.56
T1	MedQA	Decentralized	9	67.2	[61.2, 72.7]	0.75
T1	MedQA	Independent	1	66.4	[60.3, 72.0]	0.05
T1	MedQA	Independent	3	66.8	[60.7, 72.3]	0.15
T1	MedQA	Independent	5	66.8	[60.7, 72.3]	0.25
T1	MedQA	Independent	7	66.4	[60.3, 72.0]	0.35
T1	MedQA	Independent	9	67.2	[61.2, 72.7]	0.45
T1	MedQA	Self-consist.	1	68.4	[62.4, 73.8]	0.05
T1	MedQA	Self-consist.	3	65.6	[59.5, 71.2]	0.10
T1	MedQA	Self-consist.	5	65.6	[59.5, 71.2]	0.14
T1	MedQA	Self-consist.	9	66.0	[59.9, 71.6]	0.26
T1	MedQA	Self-consist.	15	65.6	[59.5, 71.2]	0.40
T1	MedQA	Hybrid	1	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	3	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	5	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	7	65.6	[59.5, 71.2]	0.05
T1	MedQA	Hybrid	9	65.6	[59.5, 71.2]	0.05
T1	MedQA	SAS zero-shot	1	66.4	[60.3, 72.0]	0.03
T1	MedXpertQA	Centralized	1	19.2	[14.8, 24.5]	0.17
T1	MedXpertQA	Centralized	3	16.0	[12.0, 21.1]	0.28
T1	MedXpertQA	Centralized	5	16.0	[12.0, 21.1]	0.40
T1	MedXpertQA	Centralized	7	17.6	[13.4, 22.8]	0.52
T1	MedXpertQA	Centralized	9	14.8	[10.9, 19.7]	0.62
T1	MedXpertQA	SAS CoT	1	15.6	[11.6, 20.6]	0.09
T1	MedXpertQA	Decentralized	1	16.0	[12.0, 21.1]	0.05
T1	MedXpertQA	Decentralized	3	15.2	[11.3, 20.2]	0.25

Table 12: Complete results grid (2 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T1	MedXpertQA	Decentralized	5	16.8	[12.7, 21.9]	0.50
T1	MedXpertQA	Decentralized	7	16.0	[12.0, 21.1]	0.74
T1	MedXpertQA	Decentralized	9	16.4	[12.3, 21.5]	1.05
T1	MedXpertQA	Independent	1	15.8	[12.9, 19.3]	0.05
T1	MedXpertQA	Independent	3	15.8	[12.9, 19.3]	0.16
T1	MedXpertQA	Independent	5	16.0	[12.0, 21.1]	0.27
T1	MedXpertQA	Independent	7	16.0	[12.0, 21.1]	0.38
T1	MedXpertQA	Independent	9	16.0	[12.0, 21.1]	0.48
T1	MedXpertQA	Self-consist.	1	16.0	[12.0, 21.1]	0.06
T1	MedXpertQA	Self-consist.	3	16.4	[12.3, 21.5]	0.10
T1	MedXpertQA	Self-consist.	5	16.0	[12.0, 21.1]	0.14
T1	MedXpertQA	Self-consist.	9	15.6	[11.6, 20.6]	0.27
T1	MedXpertQA	Self-consist.	15	16.0	[12.0, 21.1]	0.40
T1	MedXpertQA	Hybrid	1	18.0	[13.7, 23.2]	0.05
T1	MedXpertQA	Hybrid	3	18.0	[13.7, 23.2]	0.05
T1	MedXpertQA	Hybrid	5	18.0	[13.7, 23.2]	0.05
T1	MedXpertQA	Hybrid	7	18.0	[13.7, 23.2]	0.06
T1	MedXpertQA	Hybrid	9	18.0	[13.7, 23.2]	0.06
T1	MedXpertQA	SAS zero-shot	1	16.4	[12.3, 21.5]	0.03
T2	MedAgentsBench	Centralized	1	32.8	[27.3, 38.8]	0.62
T2	MedAgentsBench	Centralized	3	34.4	[28.8, 40.5]	0.93
T2	MedAgentsBench	Centralized	5	32.8	[27.3, 38.8]	1.25
T2	MedAgentsBench	Centralized	7	32.4	[26.9, 38.4]	1.59
T2	MedAgentsBench	Centralized	9	32.8	[27.3, 38.8]	1.91
T2	MedAgentsBench	SAS CoT	1	32.4	[26.9, 38.4]	0.19
T2	MedAgentsBench	Decentralized	1	33.6	[28.0, 39.7]	0.16
T2	MedAgentsBench	Decentralized	3	34.0	[28.4, 40.1]	0.73
T2	MedAgentsBench	Decentralized	5	33.2	[27.7, 39.3]	1.34
T2	MedAgentsBench	Decentralized	7	34.0	[28.4, 40.1]	2.11
T2	MedAgentsBench	Decentralized	9	34.4	[28.8, 40.5]	2.92
T2	MedAgentsBench	Independent	1	33.6	[28.0, 39.7]	0.16
T2	MedAgentsBench	Independent	3	34.8	[29.2, 40.9]	0.48
T2	MedAgentsBench	Independent	5	34.8	[29.2, 40.9]	0.79
T2	MedAgentsBench	Independent	7	33.2	[27.7, 39.3]	1.12
T2	MedAgentsBench	Independent	9	32.4	[26.9, 38.4]	1.44
T2	MedAgentsBench	Self-consist.	1	33.2	[27.7, 39.3]	0.18
T2	MedAgentsBench	Self-consist.	3	32.0	[26.5, 38.0]	0.50
T2	MedAgentsBench	Self-consist.	5	33.6	[28.0, 39.7]	0.83
T2	MedAgentsBench	Self-consist.	9	32.8	[27.3, 38.8]	1.49
T2	MedAgentsBench	Self-consist.	15	31.2	[25.8, 37.2]	2.47
T2	MedAgentsBench	Hybrid	1	32.4	[26.9, 38.4]	0.26
T2	MedAgentsBench	Hybrid	3	33.2	[27.7, 39.3]	0.49
T2	MedAgentsBench	Hybrid	5	34.0	[28.4, 40.1]	0.73
T2	MedAgentsBench	Hybrid	7	32.8	[27.3, 38.8]	0.94
T2	MedAgentsBench	Hybrid	9	32.8	[27.3, 38.8]	1.16
T2	MedAgentsBench	SAS zero-shot	1	33.2	[27.7, 39.3]	0.13
T2	MedQA	Centralized	1	86.4	[81.6, 90.1]	0.47
T2	MedQA	Centralized	3	88.8	[84.3, 92.1]	0.67
T2	MedQA	Centralized	5	89.6	[85.2, 92.8]	0.95
T2	MedQA	Centralized	7	88.4	[83.8, 91.8]	1.23
T2	MedQA	Centralized	9	86.0	[81.2, 89.8]	1.51
T2	MedQA	SAS CoT	1	88.0	[83.4, 91.5]	0.16
T2	MedQA	Decentralized	1	87.6	[82.9, 91.1]	0.13
T2	MedQA	Decentralized	3	88.8	[84.3, 92.1]	0.50
T2	MedQA	Decentralized	5	89.2	[84.7, 92.5]	0.86
T2	MedQA	Decentralized	7	88.8	[84.3, 92.1]	1.26
T2	MedQA	Decentralized	9	89.6	[85.2, 92.8]	1.75
T2	MedQA	Independent	1	87.6	[82.9, 91.1]	0.13
T2	MedQA	Independent	3	88.4	[83.8, 91.8]	0.39
T2	MedQA	Independent	5	89.6	[85.2, 92.8]	0.65
T2	MedQA	Independent	7	87.6	[82.9, 91.1]	0.90
T2	MedQA	Independent	9	88.8	[84.3, 92.1]	1.16

Table 13: Complete results grid (3 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T2	MedQA	Self-consist.	1	84.4	[79.4, 88.4]	0.15
T2	MedQA	Self-consist.	3	86.8	[82.0, 90.4]	0.41
T2	MedQA	Self-consist.	5	88.0	[83.4, 91.5]	0.67
T2	MedQA	Self-consist.	9	87.6	[82.9, 91.1]	1.21
T2	MedQA	Self-consist.	15	88.4	[83.8, 91.8]	2.01
T2	MedQA	Hybrid	1	87.6	[82.9, 91.1]	0.17
T2	MedQA	Hybrid	3	87.2	[82.5, 90.8]	0.26
T2	MedQA	Hybrid	5	88.8	[84.3, 92.1]	0.35
T2	MedQA	Hybrid	7	87.2	[82.5, 90.8]	0.43
T2	MedQA	Hybrid	9	87.6	[82.9, 91.1]	0.53
T2	MedQA	SAS zero-shot	1	87.2	[82.5, 90.8]	0.09
T2	MedXpertQA	Centralized	1	31.4	[27.5, 35.6]	0.73
T2	MedXpertQA	Centralized	3	36.0	[31.9, 40.3]	1.02
T2	MedXpertQA	Centralized	5	35.0	[30.9, 39.3]	1.38
T2	MedXpertQA	Centralized	7	36.4	[30.7, 42.5]	1.80
T2	MedXpertQA	Centralized	9	37.2	[31.4, 43.3]	2.17
T2	MedXpertQA	SAS CoT	1	32.0	[26.5, 38.0]	0.20
T2	MedXpertQA	Decentralized	1	34.0	[28.4, 40.1]	0.19
T2	MedXpertQA	Decentralized	3	38.0	[32.2, 44.2]	0.91
T2	MedXpertQA	Decentralized	5	38.0	[32.2, 44.2]	1.66
T2	MedXpertQA	Decentralized	7	38.0	[32.2, 44.2]	2.46
T2	MedXpertQA	Decentralized	9	37.2	[31.4, 43.3]	3.28
T2	MedXpertQA	Independent	1	31.8	[27.9, 36.0]	0.19
T2	MedXpertQA	Independent	3	32.6	[28.6, 36.8]	0.56
T2	MedXpertQA	Independent	5	33.8	[29.8, 38.1]	0.93
T2	MedXpertQA	Independent	7	32.8	[28.8, 37.0]	1.30
T2	MedXpertQA	Independent	9	32.6	[28.6, 36.8]	1.67
T2	MedXpertQA	Self-consist.	1	32.8	[27.3, 38.8]	0.20
T2	MedXpertQA	Self-consist.	3	36.0	[30.3, 42.1]	0.57
T2	MedXpertQA	Self-consist.	5	34.8	[29.2, 40.9]	0.94
T2	MedXpertQA	Self-consist.	9	35.2	[29.5, 41.3]	1.70
T2	MedXpertQA	Self-consist.	15	34.4	[28.8, 40.5]	2.81
T2	MedXpertQA	Hybrid	1	33.3	[27.8, 39.4]	0.29
T2	MedXpertQA	Hybrid	3	34.4	[28.8, 40.5]	0.60
T2	MedXpertQA	Hybrid	5	35.2	[29.5, 41.3]	0.89
T2	MedXpertQA	Hybrid	7	33.6	[28.0, 39.7]	1.18
T2	MedXpertQA	Hybrid	9	33.2	[27.7, 39.3]	1.56
T2	MedXpertQA	SAS zero-shot	1	32.4	[26.9, 38.4]	0.15
T3	MedAgentsBench	Centralized	1	44.8	[38.8, 51.0]	1.54
T3	MedAgentsBench	Centralized	3	44.4	[38.4, 50.6]	2.40
T3	MedAgentsBench	Centralized	5	44.8	[38.8, 51.0]	3.51
T3	MedAgentsBench	Centralized	7	46.0	[39.9, 52.2]	4.60
T3	MedAgentsBench	Centralized	9	45.6	[39.5, 51.8]	5.76
T3	MedAgentsBench	SAS CoT	1	44.4	[38.4, 50.6]	0.68
T3	MedAgentsBench	Decentralized	1	44.4	[38.4, 50.6]	0.56
T3	MedAgentsBench	Decentralized	3	44.8	[38.8, 51.0]	2.11
T3	MedAgentsBench	Decentralized	5	45.2	[39.1, 51.4]	3.88
T3	MedAgentsBench	Decentralized	7	46.4	[40.3, 52.6]	5.99
T3	MedAgentsBench	Decentralized	9	46.8	[40.7, 53.0]	7.80
T3	MedAgentsBench	Independent	1	44.4	[38.4, 50.6]	0.55
T3	MedAgentsBench	Independent	3	44.4	[38.4, 50.6]	1.61
T3	MedAgentsBench	Independent	5	44.0	[38.0, 50.2]	2.67
T3	MedAgentsBench	Independent	7	43.6	[37.6, 49.8]	3.72
T3	MedAgentsBench	Independent	9	44.8	[38.8, 51.0]	4.77
T3	MedAgentsBench	Self-consist.	1	43.6	[37.6, 49.8]	0.62
T3	MedAgentsBench	Self-consist.	3	44.4	[38.4, 50.6]	1.73
T3	MedAgentsBench	Self-consist.	5	44.4	[38.4, 50.6]	2.84
T3	MedAgentsBench	Self-consist.	9	44.0	[38.0, 50.2]	5.12
T3	MedAgentsBench	Self-consist.	15	44.4	[38.4, 50.6]	8.43
T3	MedAgentsBench	Hybrid	1	46.0	[39.9, 52.2]	0.62
T3	MedAgentsBench	Hybrid	3	45.2	[39.1, 51.4]	0.78
T3	MedAgentsBench	Hybrid	5	44.8	[38.8, 51.0]	0.99

Table 14: Complete results grid (4 of 4). All 180 multi-agent configurations and their single-model controls.

Tier	Benchmark	Architecture	n_a	Acc. (%)	95% CI	USD/1k
T3	MedAgentsBench	Hybrid	7	45.2	[39.1, 51.4]	1.09
T3	MedAgentsBench	Hybrid	9	45.6	[39.5, 51.8]	1.25
T3	MedAgentsBench	SAS zero-shot	1	46.0	[39.9, 52.2]	0.39
T3	MedQA	Centralized	1	93.2	[89.4, 95.7]	1.07
T3	MedQA	Centralized	3	91.6	[87.5, 94.4]	2.00
T3	MedQA	Centralized	5	92.4	[88.4, 95.1]	2.84
T3	MedQA	Centralized	7	92.4	[88.4, 95.1]	3.80
T3	MedQA	Centralized	9	92.4	[88.4, 95.1]	4.66
T3	MedQA	SAS CoT	1	91.6	[87.5, 94.4]	0.57
T3	MedQA	Decentralized	1	92.8	[88.9, 95.4]	0.44
T3	MedQA	Decentralized	3	92.4	[88.4, 95.1]	1.50
T3	MedQA	Decentralized	5	92.8	[88.9, 95.4]	2.62
T3	MedQA	Decentralized	7	92.4	[88.4, 95.1]	3.70
T3	MedQA	Decentralized	9	92.4	[88.4, 95.1]	4.70
T3	MedQA	Independent	1	92.4	[88.4, 95.1]	0.43
T3	MedQA	Independent	3	91.6	[87.5, 94.4]	1.29
T3	MedQA	Independent	5	91.6	[87.5, 94.4]	2.15
T3	MedQA	Independent	7	91.6	[87.5, 94.4]	3.01
T3	MedQA	Independent	9	92.0	[88.0, 94.8]	3.86
T3	MedQA	Self-consist.	1	92.0	[88.0, 94.8]	0.52
T3	MedQA	Self-consist.	3	92.8	[88.9, 95.4]	1.41
T3	MedQA	Self-consist.	5	93.2	[89.4, 95.7]	2.30
T3	MedQA	Self-consist.	9	94.0	[90.3, 96.3]	4.15
T3	MedQA	Self-consist.	15	93.6	[89.9, 96.0]	6.78
T3	MedQA	Hybrid	1	93.6	[89.9, 96.0]	0.43
T3	MedQA	Hybrid	3	93.2	[89.4, 95.7]	0.48
T3	MedQA	Hybrid	5	93.2	[89.4, 95.7]	0.52
T3	MedQA	Hybrid	7	93.2	[89.4, 95.7]	0.56
T3	MedQA	Hybrid	9	93.2	[89.4, 95.7]	0.60
T3	MedQA	SAS zero-shot	1	94.4	[90.8, 96.6]	0.27
T3	MedXpertQA	Centralized	1	46.8	[40.7, 53.0]	1.53
T3	MedXpertQA	Centralized	3	49.2	[43.1, 55.4]	2.48
T3	MedXpertQA	Centralized	5	48.4	[42.3, 54.6]	3.68
T3	MedXpertQA	Centralized	7	46.0	[39.9, 52.2]	4.89
T3	MedXpertQA	Centralized	9	46.4	[40.3, 52.6]	6.07
T3	MedXpertQA	SAS CoT	1	41.6	[35.7, 47.8]	0.72
T3	MedXpertQA	Decentralized	1	45.6	[39.5, 51.8]	0.58
T3	MedXpertQA	Decentralized	3	47.6	[41.5, 53.8]	2.33
T3	MedXpertQA	Decentralized	5	48.0	[41.9, 54.2]	4.39
T3	MedXpertQA	Decentralized	7	46.8	[40.7, 53.0]	6.65
T3	MedXpertQA	Decentralized	9	46.0	[39.9, 52.2]	9.14
T3	MedXpertQA	Independent	1	41.6	[37.4, 46.0]	0.59
T3	MedXpertQA	Independent	3	46.4	[40.3, 52.6]	1.71
T3	MedXpertQA	Independent	5	47.2	[41.1, 53.4]	2.84
T3	MedXpertQA	Independent	7	46.0	[39.9, 52.2]	3.97
T3	MedXpertQA	Independent	9	46.4	[40.3, 52.6]	5.11
T3	MedXpertQA	Self-consist.	1	42.4	[36.4, 48.6]	0.67
T3	MedXpertQA	Self-consist.	3	43.2	[37.2, 49.4]	1.83
T3	MedXpertQA	Self-consist.	5	43.2	[37.2, 49.4]	3.00
T3	MedXpertQA	Self-consist.	9	46.0	[39.9, 52.2]	5.41
T3	MedXpertQA	Self-consist.	15	47.6	[41.5, 53.8]	8.87
T3	MedXpertQA	Hybrid	1	45.2	[39.1, 51.4]	0.61
T3	MedXpertQA	Hybrid	3	44.4	[38.4, 50.6]	0.71
T3	MedXpertQA	Hybrid	5	44.8	[38.8, 51.0]	0.87
T3	MedXpertQA	Hybrid	7	44.4	[38.4, 50.6]	0.93
T3	MedXpertQA	Hybrid	9	44.4	[38.4, 50.6]	1.20
T3	MedXpertQA	SAS zero-shot	1	43.2	[37.2, 49.4]	0.46

Table 15: Capability ladder, measured on 40 items per benchmark with single-doctor CoT.

Tier	Model	Effort	MedXpertQA	MedQA	USD / 1k items
T1	`gpt-4.1-nano`	—	25.0%	75.0%	0.09
T2	`gpt-5-nano`	low	37.5%	95.0%	0.17
T3	`gpt-5-mini`	low	55.0%	97.5%	0.64

Table 16: Agent heterogeneity at n_a=3 on MedXpertQA. Strong is `gpt-5-mini`, weak is `gpt-4.1-nano`, matching Fig. 11. "Span retained" is the fraction of the all-strong minus all-weak range that the configuration preserves. Decentralized coordination has no attending, so those conditions are structurally undefined.

Capability placement	Centralized		Decentralized	
	accuracy	span retained	accuracy	span retained
All strong	49.2%	100%	47.6%	100%
Weak attending, strong specialists	45.6%	**90%**	—	—
Strong attending, weak specialists	34.4%	**57%**	—	—
Mixed-capability panel	40.8%	76%	44.0%	88%
All weak	14.8%	0%	17.2%	0%

Table 17: Collaboration gain by clinical task type on MedXpertQA, for the two tiers inside the window. Gain is the best panel configuration over the single doctor on the same items; p from paired McNemar.

Tier	Task type	n	single doctor	best panel	gain
`gpt-5-nano`	Diagnosis	84	33.3%	42.9%	+9.52 pp**
	Treatment	84	35.7%	42.9%	+7.14 pp
	Basic Science	82	26.8%	34.1%	+7.32 pp
`gpt-5-mini`	Diagnosis	84	38.1%	48.8%	+10.71 pp**
	Treatment	84	44.0%	52.4%	+8.33 pp*
	Basic Science	82	42.7%	51.2%	+8.54 pp*

$^{*}p < 0.05$, $^{**}p < 0.01$